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Kagawasan Avenue. Dumaguete City, Negros Oriental, Philippines 6200
Phone: 522-5050 Fax: 225-4751 Email: president.office@norsu.edu.ph Website: www.norsu.edu.ph

ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT

A Capstone Project

Presented to the Faculty of the

College of Arts and Sciences

Computer Science and Information Technology Department

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In Partial Fulfillment of the Requirement

for the Degree

Bachelor of Science in Information Technology

By:

KERWIN P. BABOR

MARY JOYCE F. CANDIDO

MYCA C. CAPINGCOT

DIONGIE C. FUNDADOR

GINO Q. MALCO

MAY 2024

APPROVAL SHEET

In partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY** (BSINT), this Capstone Project entitled: "**ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT**" prepared and submitted by **DIONGIE C. FUNDADOR; GINO Q. MALCO; KERWIN P. BABOR; MARY JOYCE F. CANDIDO;** and **MYCA C. CAPINGCOT,** has been examined and is recommended for oral defense.

SHERAMIE C. REYES
English Critic

MA. ARCHEL L. CALUNSAG
Project Adviser

APPROVED by the **PANEL of EVALUATORS** for **ORAL DEFENSE** on May 2, 2024 with a grade of _____.

MR. MARK ANTHONY A. CATALUÑA
Member

MR. MICHEAL C. YLEAÑA
Member

MR. JETTY A. CASIDO
Chairperson

ACCEPTED and **APPROVED** in partial fulfillment of the requirements for the Degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY.**

ZENAIDA D. CALUMPANG, MSIT, Ed.D.
Dean, College of Arts and Sciences

DEDICATION

This work is dedicated to our family and friends, whose unwavering support and encouragement have been the foundation of our success. We are incredibly grateful to our mentors and professors for not just your invaluable guidance and inspiration, but also for shaping our understanding and approach to this project. Finally, to our colleagues and everyone who believes in the potential of this project, your faith has been a driving force in bringing this vision to life. And above all, we dedicate this work to God for providing the strength and wisdom to persevere. Thank you all for being part of this incredible journey.

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ABSTRACT

Title:	**ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT**
Author:	**BABOR, P.; CAPINGCOT, C.; CANDIDO, F.; FUNDADOR, C.,** and **MALCO, Q**
Degree:	**BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**
School:	**NEGROS ORIENTAL STATE UNIVERSITY**
Year:	**2023-2024**

Implementing Cugon Bamboo Resort's Online Reservation System is a strategic move that promises to enhance operational efficiency and visitor satisfaction. By consolidating numerous resort services into a single digital platform, this system offers secure user authentication, streamlined reservation administration, simplified check-in/out processes, automated billing, and efficient handling of guest service requests. This technological advancement aligns the resort with the evolving digital landscape and fosters seamless communication between staff and guests. It not only bolsters visitor loyalty and expands the resort's customer base but also ensures our guests feel comfortable and confident in their interactions with the new system.

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CHAPTER I

INTRODUCTION

Project Context

In today's digital age, access to information is virtually at our fingertips. The rapid digitalization of the business landscape has led to nearly every company or establishment establishing an online presence through websites or dedicated online platforms. Our Online Reservation System for Cugon Bamboo Resort allows staff and guest services to communicate quickly and efficiently. Its main goal is to make our operations more efficient so that our team can spend more time serving guests at the reception desk. By implementing this system, we anticipate significantly reducing administrative tasks, leading to increase in staff productivity. It will result in a cost savings annually. The main goal is to create a comprehensive resort management system that will allow us to efficiently oversee our resort functions, including reservations, guest services, staff management, and more.

The researchers created a system to meet two needs: to keep up with the changing digital world and improve the overall guest experience. Technology is not just a tool but a strategic necessity for staying relevant and efficient in today's competitive hospitality industry. By introducing this system, researchers are not only keeping up with the digital age but also positioning themselves at the forefront of it. It is a strategic move, a decision that will keep us connected with our guests, provide them with a seamless experience from booking to check-out, and ultimately build their loyalty (Aiman Balowa A. et al. 2017).

This system offers a range of essential functionalities designed to streamline resort operations. It includes robust user authentication and access control features, ensuring secure access for administrators, staff, and managers. Our system is built with the latest security protocols, ensuring the safety of guest and staff information. The reservation management module is user-friendly, allowing staff to search for rooms based on guest preferences and facilitating reservation creation, modification, and cancellation, all through a user-friendly booking calendar. Check-in and check-out processes are simplified, and billing is automated. Guests can request additional services through the system, while staff can efficiently manage and track these

requests. Staff management is also optimized with records and task management features, enhancing overall resort efficiency.

Objectives of the problem

The aim of this study is to introduce an electronic reservation system at Cugon Bamboo Resort, a significant upgrade from their current manual system. This transition promises to enhance the efficiency of their transactions, improve security measures, and provide a reliable backup system for all successful transactions.

- To increase the security between the management and staff
- To lessen time consume
- To create a backup of files for every successful transaction.
- To make it easier to find the availability of the facilities and amenities.

Scope and Limitation

The scope of this research is to explore the programming languages used in the development, benefits, and potential drawbacks of using online reservation systems:

Scope

This study covers markup languages such as HTML to build the base foundation of the website, CSS for styling, and two programming languages such as JavaScript to create dynamic content for the website and PHP/MySQL for the database. Enhancing customer convenience often arises from incorporating specific technologies or simplifying procedures, ensuring a smoother and more user-friendly experience. Simultaneously, Cugon Bamboo Resort benefits from reduced costs by implementing efficient technologies and optimizing its operational processes, leading to significant savings. Improved efficiency is a natural byproduct of technological advancements and streamlined business processes, fostering a more productive and competitive environment. Moreover, leveraging technology allows Cugon Bamboo Resort to reach a wider audience, extending its outreach to potential customers through various channels.

Limitation

The study faces several limitations. Firstly, the system will offer a GCash payment method through email confirmations but will not support a direct API gateway or other payment options such as PayPal and others, potentially limiting payment flexibility for users. Secondly, there are potential limitations related to sample size and data quality when evaluating online reservation systems. The research might be constrained by a small sample size, potentially limiting the generalizability of the findings to a broader context. Additionally, the quality of the data collected could vary among the online reservation systems under examination, posing challenges to the overall reliability and validity of the study. Data quality assessment encompasses considerations such as accuracy, completeness, and consistency, factors that can influence the robustness and credibility of the research outcomes. Thus, careful interpretation of the results is warranted, considering these inherent limitations in sample size and data quality.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter thoroughly reviews related literature and studies about online resort reservation systems. The focus is on how this system can significantly benefit Cugon Bamboo Resort, which is eager to transition from manual reservations to an automated online system, thereby enhancing its operations and customer service.

The literature review commences by noting the prevalent use of technology in the Philippines' hospitality industry. While larger resorts have readily embraced online reservation systems (ORS), it is worth highlighting that some remote provinces and smaller establishments still rely on manual processes, such as signage, print advertisements, and logbooks for customer data. This stark contrast underscores the competitive advantage that Cugon Bamboo Resort can gain by considering the benefits of transitioning to an online system and deciding to transition to a confident and reassuring one.

"Hotel Reservation System" has revolutionized the Philippine hospitality industry by enabling hotels and resorts to streamline operations and improve customer service. However, not all establishments have transitioned to this technology. ORS are computerized platforms that store and disseminate information about hotels and other lodging facilities. They facilitate efficient online marketing and sales management, empowering hoteliers to upload room rates and availability information, which can be instantly accessed by various sales channels, simplifying the intricate task of reaching a wider audience (Agilysys, 2023).

Customers are increasingly drawn to online reservations due to their convenience and security. They can now book rooms from the comfort of their homes, ensuring the safety of their personal and financial information. Furthermore, online travel agents provide a platform for travelers to compare prices and amenities across various hotels, thereby enhancing their decision-making process. These booking systems offer a wide range of benefits, including enhanced organization, security, efficiency, and customer service. They are a crucial tool in today's digital age, streamlining processes and significantly improving the overall customer experience, a

factor that Cugon Bamboo Resort's management and decision-makers should consider with empathy and consideration.

The booking system involves systematically registering guest names and relevant information for check-ins and using hotel facilities and services. It is a pivotal resource management and booking tool for customers and hotel operations, ensuring that services are efficiently delivered and reservations are accurately recorded. They can rapidly process information and enable real-time adjustments, perform the same tasks consistently without fatigue or error, follow precise instructions meticulously, operate in hazardous environments, work consistently without breaks, holidays, or additional expenses, and be reprogrammed to handle different tasks.

The integration of online reservation systems has not only revolutionized but also significantly transformed the Philippine hospitality sector. By replacing manual processes with streamlined, efficient, and customer-focused technologies, these systems have paved the way for a new era of hospitality. They offer a wealth of advantages, ensuring that the industry remains at the forefront of technological advancements and enhanced customer service.

Related Studies

International

Odemba's (2019) study on the rise of web services in distributed computing, mainly the internet's text of the hospitality industry, significantly contributes to our understanding of the transition from data-oriented to service-oriented architectures. The research culminates in developing an online reservation system that is accessible as a web service for other hospitality providers, a finding that is relevant to our field. By focusing on Simple Objects Access Protocol or SOAP for its robust standards and tools, the study identifies the technological aspects and business benefits of web service implementation, a crucial insight for our professional peers.

Odemba emphasizes web service applications in the hospitality industry, resulting in the development of an online reservation system that is accessible as a web service for fellow hospitality providers. Similarly, our study concentrates on the adoption of web-based reservation systems. In contrast, Odemba's study centers on SOAP for its robust standards and tools, which can also be applied in ours, that align

the exploration of both the technological aspects and business benefits associated with implementing web services in the context of reservation systems.

National

The study of Zhao Y et al. (2022) entitled “Online Reservation Intention of Tourist Attractions” investigated the influence of subjective norms, government policy, perceived usefulness, perceived ease of use, and perceived risk on reservation intentions of tourist attractions. An online survey was conducted in China, and 255 questionnaires were collected. The data were analyzed using SPSS 26.0 and AMOS 28.0 to construct a structural equation modeling and explore the path. The findings show that (1) subjective norms have no significant impact on reservation behaviors under voluntary situations; (2) perceived usefulness positively affects tourists’ reservation intention; and (3) perceived risk has a significant negative impact on reservation intention, and government policy is the main factor affecting tourists’ reservation intentions. These findings enhance understanding of tourists’ reservation intentions and extend the TAM theory.

Despite the different contexts, both the Chinese study and our research share a common approach to understanding what drives booking decisions. They both examine social influences, government policies, and user perceptions, highlighting the importance of these factors in shaping reservation behavior. This shared understanding can foster a sense of collaboration and mutual learning among our academic peers, even if we are operating in different industries or settings.

Local

M. J. Garcia-Granja, et. al (2022). "Development of an Innovative Seat Reservation System for University Buildings Based on BIM Technology." This study aims to show the benefits of using Building Information Modeling (BIM) to create a real-time system for managing seat occupancy. Using BIM, this system can provide personalized reservation solutions, giving users various ways to access model information and make the best seating choices. It is a powerful tool that can be used for multiple purposes, including facilities management. However, the use of BIM for facilities management in university spaces has yet to be widely adopted. In university

spaces, it is often difficult to check the availability of seats, especially in public facilities with free access. It can lead to capacity problems. Additionally, while BIM can be used to create attractive and intuitive graphic representations of buildings and facilities, there are still challenges regarding how to make this information accessible to the public in a way that takes into account their preferences.

Our reservation system study and the essay share common ground in exploring reservation systems, but their technology approaches differ.

Both studies emphasize maximizing the utilization of available resources, notably seats, via an innovative reservation system. While our research focuses on general reservation management, the article narrows its focus to the university setting, emphasizing the difficulties in monitoring seat occupancy in such settings.

Conceptual Framework of the Study

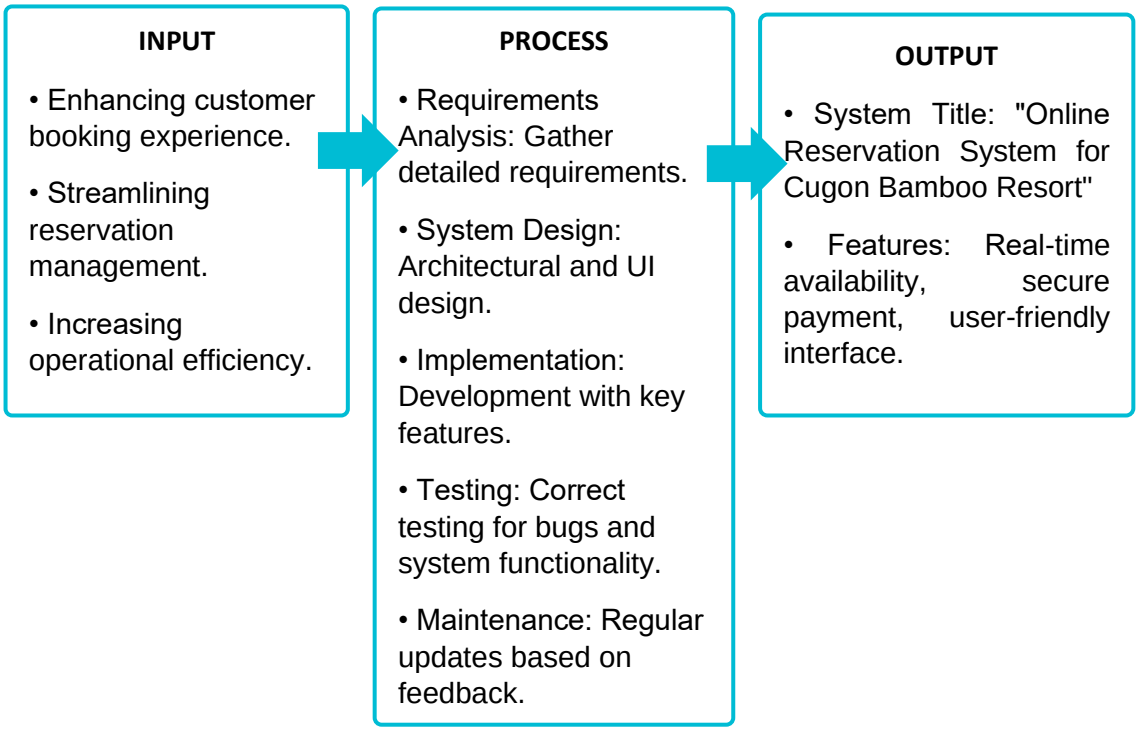

Figure 1. Conceptual Framework of the Study

The conceptual framework diagram of the 'Online Reservation System for Cugon Bamboo Resort' is based on the Input-Process-Output (IPO) Model. This model is a structured representation of the system's essential parts and processes. It is divided into three major sections: input, processing, and output, each addressing a

different component of system development. The IPO Model is particularly useful in understanding how the system functions and how its various components interact with each other.

The basic concepts that will lead to the system's creation are emphasized in the input phase. These suggestions included improving the consumer booking experience by making it easier to understand and user-friendly, streamlining the reservation administration process to make it more effective for the resort's staff, and increasing operational efficiency to minimize expenses and improve service delivery. This phase is critical because it defines the system's direction and objectives, ensuring that the development meets the specific needs and problems of Cugon Bamboo Resort. The system's implementation is expected to result in [insert specific benefits here], which will significantly enhance the resort's operations and customer service.

For the system's development, we employed the waterfall model of the Software Development Life Cycle (SDLC), known for its thoroughness and reliability. This model involves a comprehensive requirements analysis, system design, implementation, testing, and maintenance. Its linear and sequential approach is particularly effective for managing large and complex projects, ensuring the system's quality and reliability.

The final output of this process is the 'Online Reservation System for Cugon Bamboo Resort.' This system is characterized by key features such as real-time availability checks and secure online payment choices, all seamlessly integrated within a user-friendly interface. This interface is designed to make the audience feel at ease with the new technology, immediately addressing the inputs by enhancing the booking experience and streamlining operational operations.

Theoretical Framework of the Study

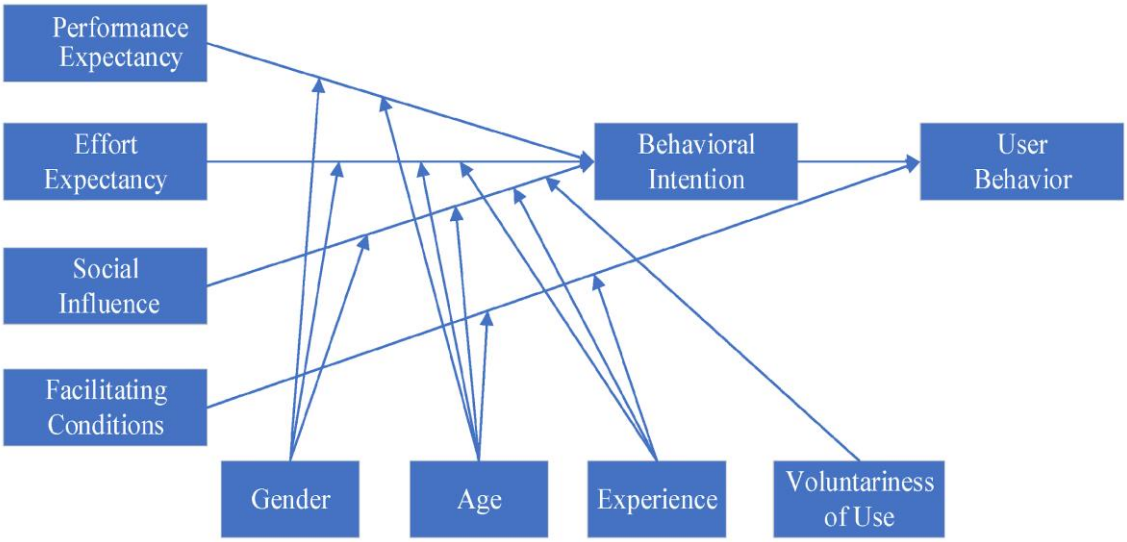

Figure 2. Theoretical Framework of the Study

The UTAUT Model, or Unified Theory of Acceptance and Use of Technology, is a comprehensive framework for explaining user perceptions and acceptance behavior in technology adoption (Venkatesh et al., 2003). According to this theoretical model, actual technology usage is influenced by behavioral intention, which is, in turn, determined by the direct impact of four critical constructs: performance expectancy, effort expectancy, social influence, and facilitating conditions. Additionally, the influence of these predictors is moderated by individual factors such as age, gender, experience, and voluntariness of use (Venkatesh et al., 2003).

The UTAUT model, based on various technology acceptance theories, focuses on performance expectancy, which is the belief that using a system will improve job performance. This expectancy is a strong predictor of usage intention in both voluntary and mandatory settings. Effort expectancy, which measures the ease of system use, diminishes after prolonged use. Social influence, facilitating conditions, and moderating factors like age, gender, experience, and voluntariness of use also influence intention. UTAUT demonstrates significant predictive power, accounting for 70% of the variance in use intention, emphasizing the complexities of personal and demographic factors in the technology acceptance process.

Applying the UTAUT model to the Cugon Bamboo Resort's Online Reservation System can provide valuable insights into the factors influencing user acceptance and utilization of the system.

Several variables aligned with the principles of the UTAUT model contribute to a positive user experience in the context of Cugon Bamboo Resort's Online Reservation System. Performance Expectancy emphasizes that guests who use the system should regard it as improving their booking experience. This can be accomplished through an easy-to-use interface, detailed information about hotel alternatives and facilities, and promotional offers that add value to their reservations. Effort Expectancy emphasizes the necessity of developing the system for user-friendliness, ensuring an easy and convenient reservation procedure. It includes simplifying the booking process, offering clear instructions, and maintaining a responsive design across several devices.

The resort's ability to display good reviews and testimonials from delighted guests who have used the online reservation system successfully leverages social influence. The resort can boost its perceived value by promoting the system on social media and emphasizing its popularity among other visitors. The Facilitating Conditions underline the importance of guests believing in the resort's organizational and technological assistance for the online reservation system. This includes ensuring the system's dependability, security, and seamless integration with the broader customer service infrastructure.

Consideration of Moderating Factors such as age, gender, experience, and voluntariness of use is critical for accommodating the resort's different populations. For example, if the resort attracts people of all ages, the system should be simple. Gender preferences and user experience levels should also be considered in the design and promotion of the system. When reservations are required, such as for special packages or during peak seasons, emphasizing compliance is critical. Cugon Bamboo Resort may improve customer satisfaction, boost system adoption, and personalize the experience to the preferences and needs of its diverse clients by aligning the Online Reservation System with these concerns. Regular changes based on user feedback will improve the system's effectiveness over time.

Cugon Bamboo Resort may boost user satisfaction, encourage system adoption, and improve overall guest experience by aligning the design and promotion of the Online Reservation System with the concepts of the UTAUT model. Regular feedback and updates based on user interactions can help the system become more effective.

CHAPTER III

TECHNICAL BACKGROUND

Technicality of the Project

Our project advocates have implemented a user-friendly web-based program that allows users to interact with it through various digital platforms. The Reservation System for Cugon Bamboo Resort is specifically designed as an online website, providing accessibility to individuals utilizing electronic devices and facilitating the workflow for staff members. This modernized approach eliminates the need for traditional methods.

In the context of this project, several technical terms are employed to describe the tools and technologies utilized. For instance, we utilize VS Code as our text editor. At the same time, server-client-side components such as PHP, HTML, CSS, Apache, and databases like Wamp and MySQL contribute to the system's functionality. The web application's core functionalities are supported by these technologies, and PhpMyAdmin is a tool for managing the MySQL database.

Details of the technologies to be used

This project will be available on any technology, such as mobile phones, laptops, and computers.

In developing the project, the following technology tools were used:

VS Code — In the development of the system, the researchers used Microsoft Visual Studio Code to encode the source code of the system to be the accurate and fast acquisition of data and information.

PHP— is a general-purpose scripting language especially suited to server-side web development where PHP generally runs on a web server. It can also be used for command-line scripting and client-side GUI applications.

WAMP—Developers used this as a development tool to allow website designers and programmers to test their work on their own computers without access to the Internet.

MySQL is an open-source relational database management system. Developers will use MySQL because it is one of the system development-compatible databases.

How the project will work

The Online Reservation System for Cugon Bamboo Resort is a game-changer in adapting to the ever-changing digital world. A strong internet presence is crucial in today's highly competitive hotel sector. This system is a comprehensive resort management solution that aims to significantly improve operational efficiency while also enhancing the overall visitor experience. The primary goal is to enhance communication between staff and guest services, empowering our team to spend more time serving visitors at the front desk, thereby boosting their productivity and job satisfaction.

The strategy, designed to address the dual objectives of remaining technologically relevant while boosting visitor happiness, positions Cugon Bamboo Resort at the forefront of technical innovations in the hospitality sector. By embracing this technology, we are confident in developing a seamless relationship with our visitors, providing them with an effortless experience from reservation to check-out, thereby fostering consumer loyalty and ensuring the resort's long-term success.

The system included several critical functions that help to optimize resort operations. Thanks to robust user authentication and access control capabilities, administrators, personnel, and managers have safe access. The reservation management module enables staff to conveniently search for rooms based on visitor preferences and create, modify, and cancel reservations using a user-friendly booking calendar. Check-in and check-out operations, as well as automatic billing, simplify the entire visitor experience. Additionally, the technology allows clients to request extra services, which the staff can handle and track effectively. The system seeks to improve overall resort efficiency by providing improved staff management capabilities such as keeping records and task management. Cugon Bamboo Resort's deployment of the Online Reservation System is more than a technological improvement; it is a strategic step to stay ahead of the digital curve and provide outstanding service to our guests.

CHAPTER IV

RESEARCH METHODOLOGY

This qualitative study used observations and document analysis to understand reservation procedures, customer interactions, and organizational structures at Cugon Bamboo Resort. It also examined consumer and staff opinions on the planned online system, contributing to the creation of a user-oriented system.

Environment

Locale

The research environment outlined includes the Municipality of Bindoy, which is located in Negros Oriental, Philippines. Bindoy, formerly known as Payabon, is a third-class municipality with a population of around 40,308 people, according to the 2020 census. This location is the backdrop for developing the Online Reservation System for Cugon Bamboo Resorts.

The Online Reservation System is designed to operate digitally, allowing guests to make reservations and bookings through an online platform accessible from anywhere in the Philippines with an internet connection. This online structure enables smooth interactions between potential guests and the resort, regardless of location. The system is specifically created to meet the specific needs of Cugon Bamboo Resort, located in Pangalaycayan, Bindoy, Negros Oriental. This strategic location emphasizes the need to capture the local context and demographic concerns in the system's development and deployment, assuring its relevance and success within Bindoy's particular geographic and cultural environment.

Organizational Chart/Profile

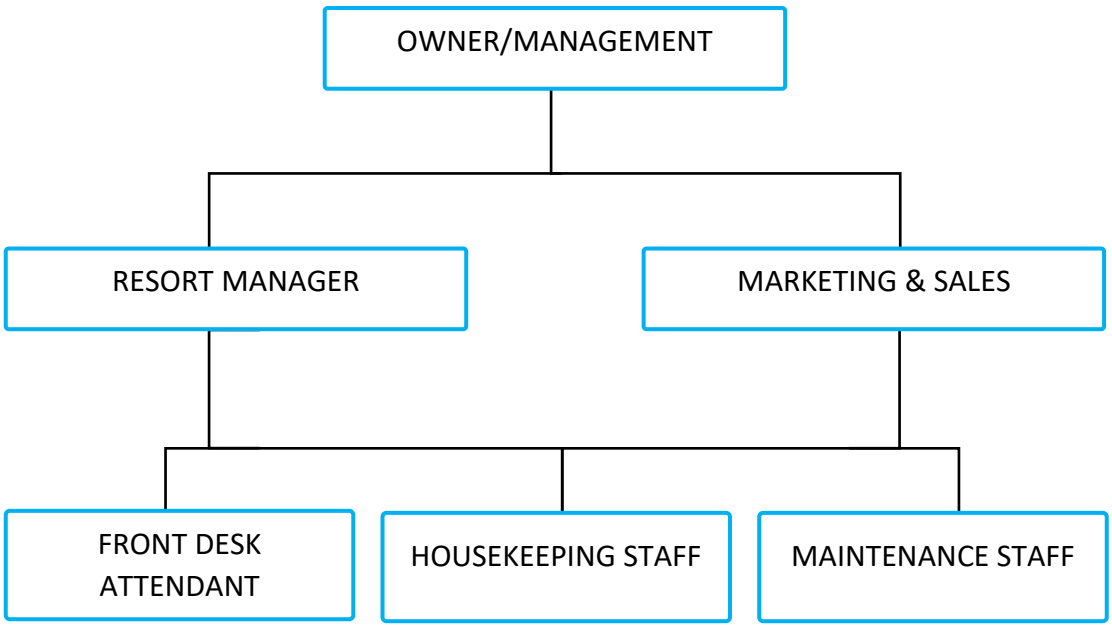

Figure 3. Organizational Chart/Profile

This figure presents Cugon Bamboo Resort's organizational chart, with a clear emphasis on the management hierarchy and structure. The Owner/Management holds the topmost position, managing the overall organization. Below this level, the structure is divided into two distinct roles: Resort Manager and Marketing and Sales. The Resort Manager, as the second-in-command, supervises the resort's day-to-day operations, supported by three major personnel roles: Front Desk Attendant, Housekeeping staff, and Maintenance Staff. On the other hand, Marketing & Sales, as a separate entity, is responsible for advertising the resort and increasing bookings. This chart not only describes the resort's duties and reporting structure but also provides a clear and orderly view of the resort's operational and managerial responsibilities, giving the audience a sense of order and clarity.

Operational Feasibility

The operational feasibility of utilizing an Online Reservation System at Cugon Bamboo Resort was evaluated using various essential criteria. The resort has adequate technological infrastructure to support the system's installation and maintenance. The resort's staff are confident about technology adoption, suggesting a willingness to use

it efficiently. The success of the system's implementation hinges on the provision of proper training and assistance, which can enable the staff to use the system successfully. This, in turn, will lead to seamless operations and excellent customer service. Given the growing trend of online bookings in the hospitality business, customers are expected to embrace the new reservation system, which offers simplicity and improves their overall booking experience.

Fishbone Diagram

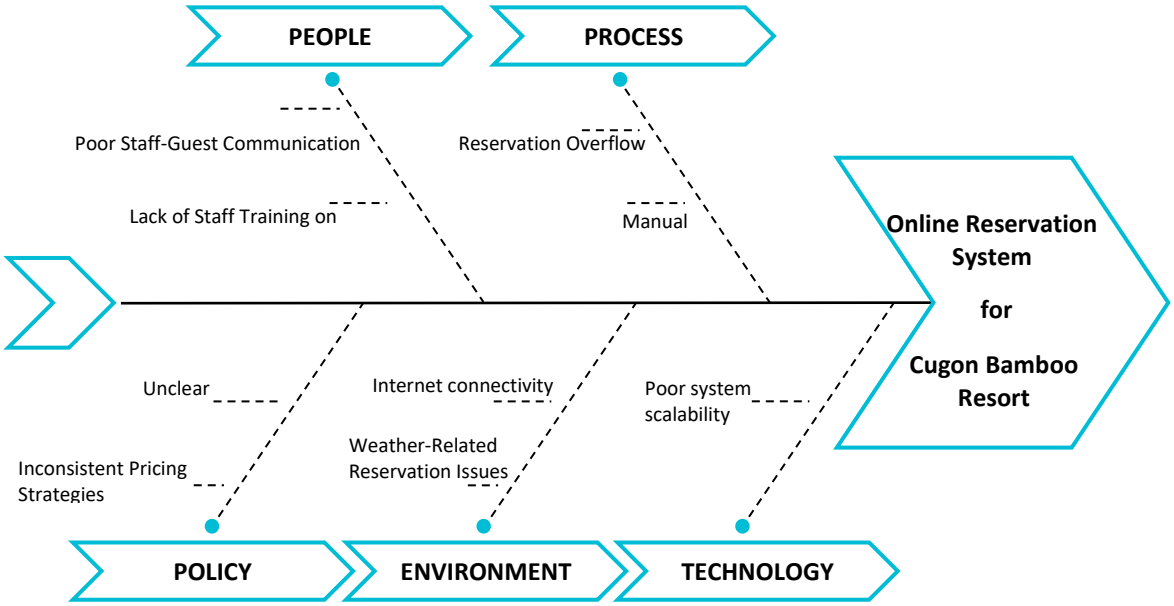

Figure 4. Fishbone Diagram

The figure above shows five key factors causing the inefficient reservation process at Cugon Bamboo Resort: People, Process, Technology, Environment, and Policy. It highlights issues like poor staff communication, untrained staff, and weather-related reservations.

Functional Decomposition Diagram

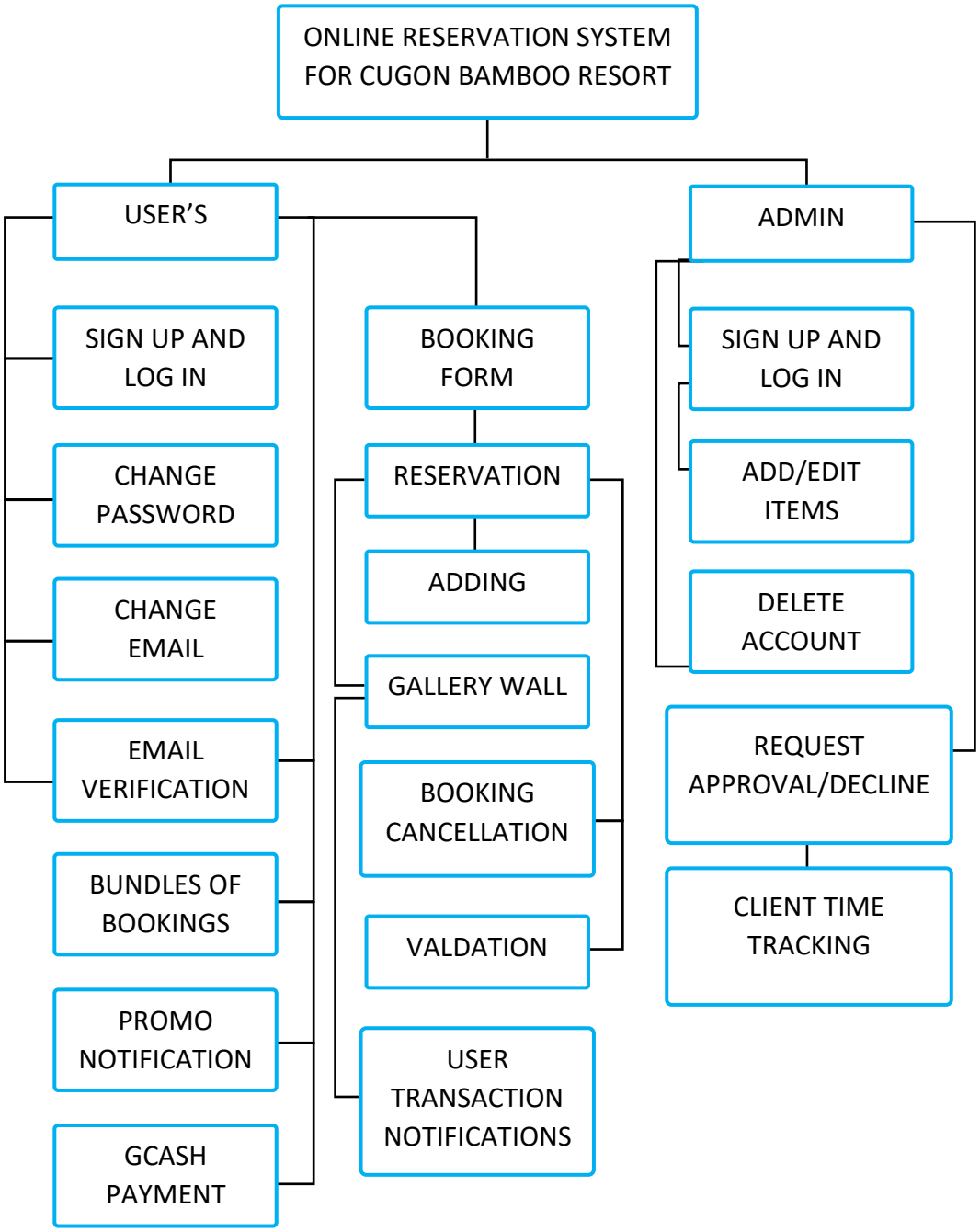

Figure 5. Functional Decomposition Diagram

The diagram above displays how the Online Reservation System works. It highlights features such as canceling bookings, sending notifications, responding to requests, and monitoring client schedules. The following outline summarizes the system's functionalities, demonstrating its ability to manage reservations, engage with users, and ensure tracking of customer interactions.

Technical Feasibility

The resort's technical feasibility is assessed to determine whether it contains the necessary hardware and software and to examine and evaluate the necessary technologies for system development. Currently, we possess [specific technology and software], which makes the suggested online reservation system and website feasible. The Cugon Bamboo Resort utilizes free and readily available technology.

Software- it will not require any special software.

Software compatibility

- The system is compatible with popular Web browsers such as Google Chrome, Firefox, Safari, and Microsoft Edge.
- It is essential to have compatibility with different operating systems like Windows, macOS, iOS, and Android.

Hardware Compatibility

- The system is compatible with various devices, such as desktops, laptops, tablets, and smartphones.
- It is compatible with different screen sizes and resolutions to ensure proper display and usability.

Other technologies compatibility

- Ensure compatibility with different internet connection speeds to provide a smooth user experience.
- The proposed system is compatible with Gcash payment, a widely used digital payment platform in the Philippines. This integration will streamline our financial operations and provide a convenient payment option for our customers.
- Compatible with web standards and protocols such as HTTP/HTTPS.

The system is compatible with security measures such as SSL/TLS encryption, which ensures that all data transmitted between the web server and browser remains encrypted and secure. It also includes cross-site scripting (XSS) prevention, a security feature that protects against malicious scripts, and protection against SQL injection attacks, a technique used to exploit vulnerabilities in a website's database layer.

By ensuring compatibility across various dimensions, the online reservation system can offer a smooth experience to consumers regardless of their hardware,

software, or other technological capabilities. This user-friendly nature of the system instills confidence in the stakeholders about its easy adoption and use.

Relevance of the Technology

An online reservation system offers a host of advantages, including the ability to support essential features, ensure security, and enhance the user experience. However, its most significant benefit lies in its scalability. This scalability not only assures the technical team of its long-term viability but also its adaptability to future needs, a crucial aspect that the team will diligently oversee.

Front-end Technologies

HTML5, jQuery, Flowbite Tailwind libraries, Bootstrap, and the Node package manager(npm) are foundational for creating the user interface, layout, and interactivity of the online reservation system.

Back-end Technologies

PHP: Determine the logic, functionality, and performance of the system.

Database Management

Relational databases (MySQL): securely store reservation data, user information, and system configurations.

The proposed web-based online reservation system offers a secure, effective, and user-friendly solution for both customers and administrators at Cugon Bamboo Resort. By automating the reservation process, it will significantly reduce the reliance on manual methods, paving the way for future growth and scalability. This system will establish an efficient information management system, making it incredibly convenient for Cugon Resort to operate. Moreover, its web-based nature means it can be accessed from anywhere, eliminating the need for a dedicated technical professional to manage the system. This will undoubtedly provide a sense of relief and ease for the management.

Schedule Feasibility

Table 1

Gantt Chart

Activity/Task	**October** 2023				**November** 2023				**January** 2024				**February** 2024					**March** 2024			**April** 2024			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Project Planning																								
Requirements Gathering																								
Title Proposal																								
System Planning/Layout																								
Development																								
Testing																								
Editing & Revisions																								
Implementation																								

The table shows the whole schedule that the researchers followed throughout the development of the study. The proponents utilized the Gantt chart to track the project's timeline for assigned responsibilities.

Economic Feasibility

Table 2

Cost-Benefit Analysis

		Item	Amount
Cost	**Hardware**	• Computer set • Printer / Scanner	₱ 25,000 ₱ 9,000
	Software	• Anti-virus, encryption software • SSL drivers' certificate • Composer for Emails	
	Monthly Hosting	• Monthly Hosting Fees	₱ 1,500
		Total	₱ 35,500
Benefit		• Streamline reservation process. • Improve operational efficiency. • Reduces administrative costs and labor costs • Enhanced booking experience attracts more guests.	

The Online Reservation System for Cugon Bamboo Resort requires an initial investment of ₱34,000 for hardware and software, plus a monthly hosting fee of ₱1,500. This investment streamlines the reservation process, boosts operational efficiency, reduces costs, and enhances the guest experience, leading to increased occupancy and revenue. Additionally, the system reduces administrative and labor costs by automating the reservation process. The long-term benefits include improved efficiency and higher income production for the resort.

Requirements Modelling

Table 3

Inputs, Process, Outputs

Inputs	Process	Output
• User Information • Booking Information • Room Information	• **Updates system inventory records** • **Generates Reports**	• **Sends Confirmation Email** • **Receipts for payment transactions** • **Room availability updates**

Input: System inputs include all data that enters the system, whether manually inputted by humans or generated automatically by other systems. The reservation system accepts user information such as name, contact information, booking dates, hotel room preferences, and special requests. Inputs may contain inventory data, such as room availability, pricing, and promotional offers.

Process: System processes are the logical rules and procedures used for turning data that enters into helpful information or outputs. The reservation system's processes include confirming guest input data to ensure accuracy and completeness, calculating total booking costs based on selected options, updating inventory records to reflect booked accommodations, and generating reports for management analysis.

Output: System outputs are information the system collects in response to user inputs or system activities. The reservation system sends confirmation emails or messages to guests confirming successful bookings, electronic receipts for payment transactions, reservation reports for resort staff, and room availability updates shown on the resort's website or mobile app.

Performance: Performance factors include system qualities such as speed, volume, capacity, availability, and stability. The reservation system should be able to handle a high amount of concurrent user requests while maintaining quick responses and no disruptions. It should also have enough capacity to keep up with rising demand during high booking periods while ensuring consistent availability to meet customer reservations efficiently.

Security & Control: Security and control requirements are crucial in securing the system and its data from external and internal attacks. It includes implementing robust authentication mechanisms to verify user identities, encryption protocols to protect sensitive data during transmission, access control measures to prevent unauthorized access to system resources, and audit trails to monitor and track system activities for compliance and accountability purposes. In addition, backup and recovery protocols should be in place to ensure data integrity and operational continuity in the case of system failures or disasters.

Data and Process Modelling

Context Diagram

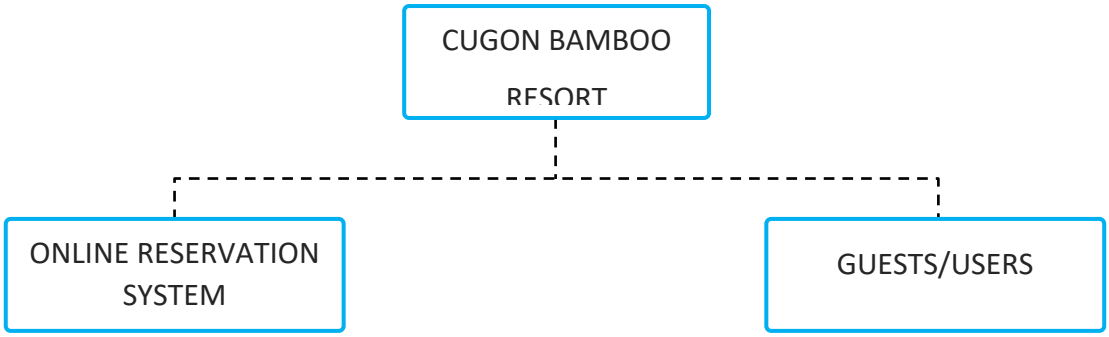

Figure 6. Context Diagram

The diagram shows how the Online Reservation System, Cugon Bamboo Resort, and users outside the resort are connected. Cugon Bamboo Resort is central to this setup, while the Online Reservation System is a tool for booking accommodations. Guests or users use this system to make reservations. It gives a summary of how the reservation system works within the resort and engages with people outside it.

Data Flow Diagram

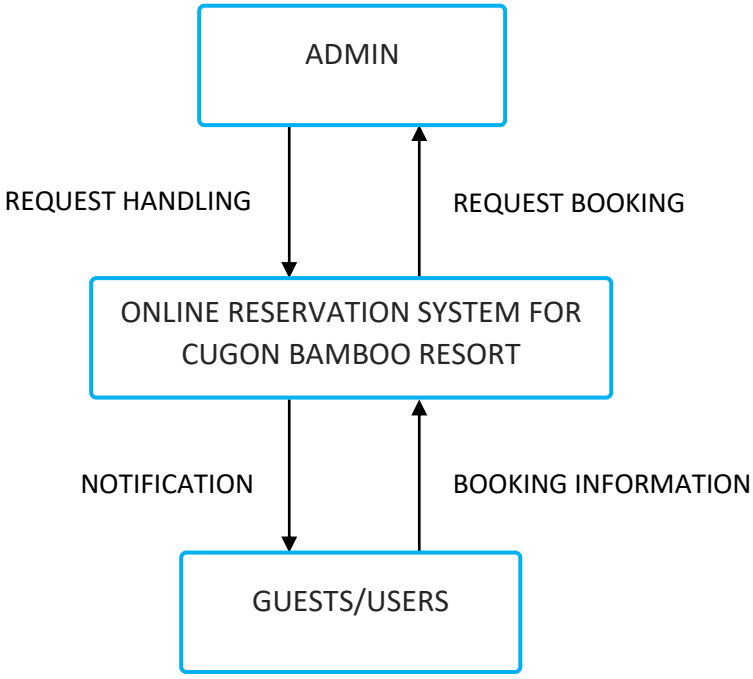

Figure 7. Data Flow Diagram

The figure above shows users outside the system interact with the Online Reservation System, where their information is handled. The diagram showcases four primary data flows; “Booking Information" (details provided by users), “Request Booking” (request approval/decline), "Request Handling" (managing user requests), and "Notifications" (messages sent to users). This summary of the Data Flow Diagram provides a brief look into how data moves through the Online Reservation System for Cugon Bamboo Resort.

System Flowchart

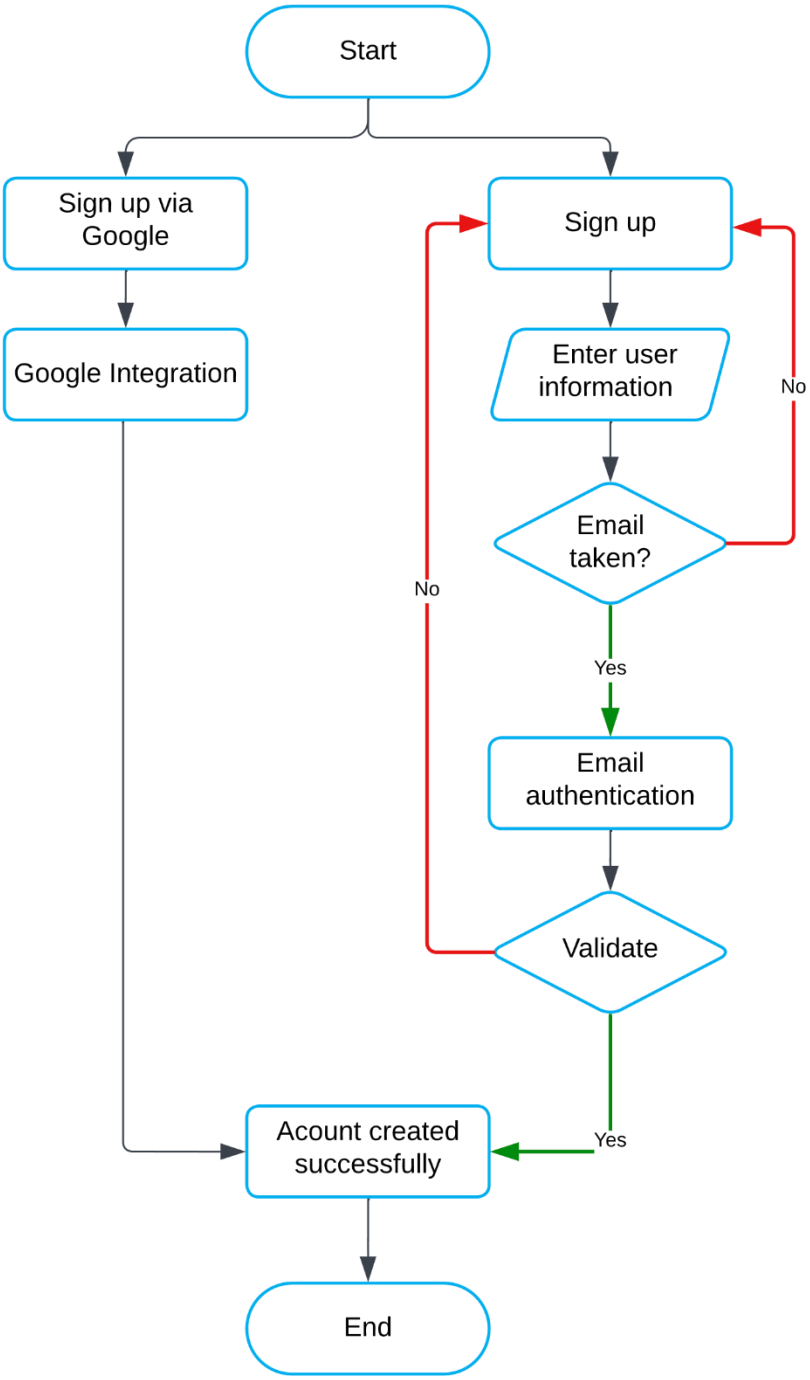

Figure 8. System Flowchart

The figure shows two sign-up options (Standard Sign-up and Google Sign-up). In Google Sign-up, users must verify their account through the Google integration form and confirm their email. Standard sign-up users enter their information through regular sign-up; it is then verified that their email address is available and runs an authentication process. Once the email is verified, the system verifies the user's details and creates the account successfully.

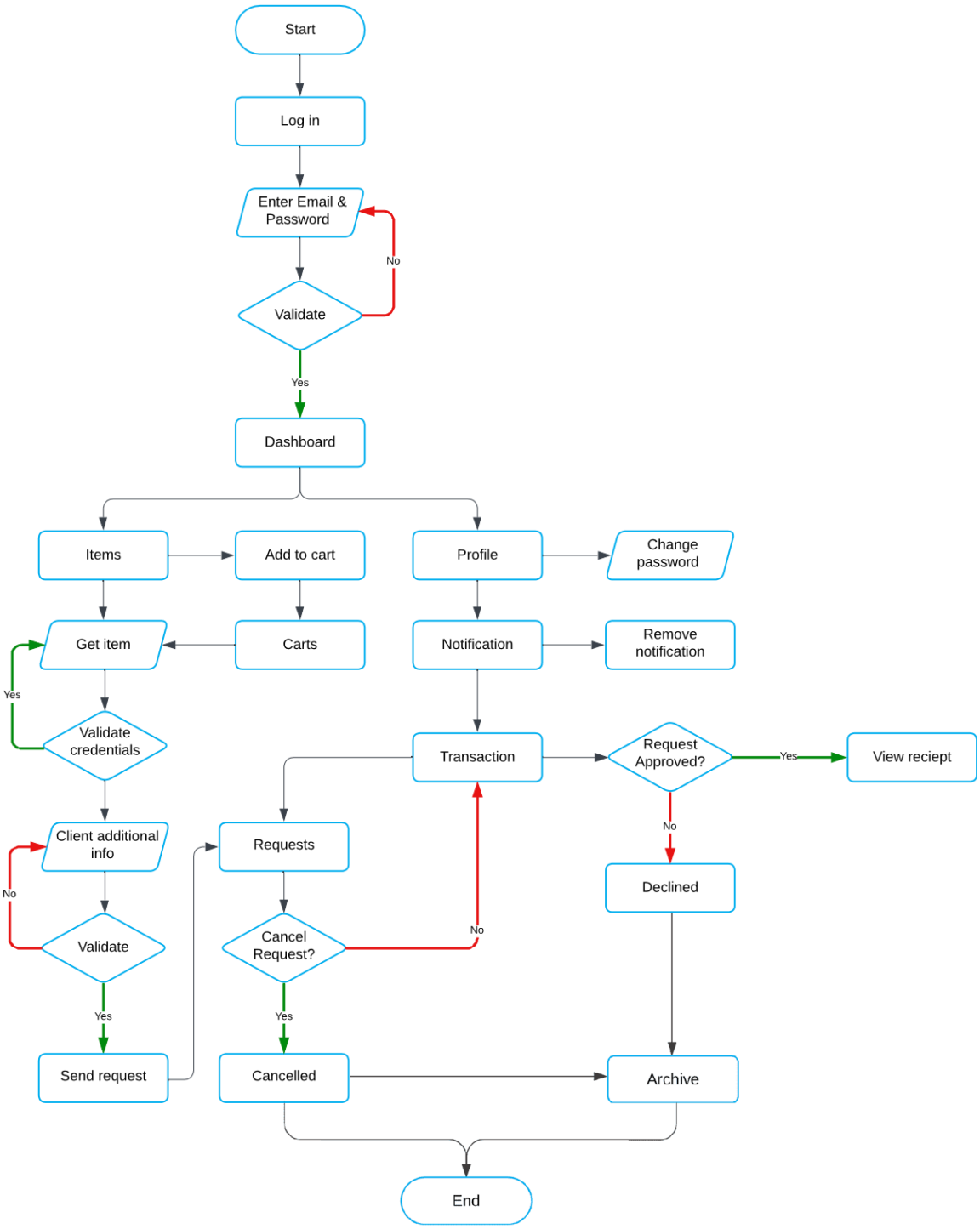

Figure 9. System Flowchart

The flowchart shows a user-friendly process that begins with logging in and inputting personal information. Users may browse their profiles and notifications, reset their passwords, examine transaction details, and track their status. Users have the option of archiving approved requests and canceling requests, which are also archived. The approach includes validating credentials, adding items to the basket, providing extra information, and sending or canceling requests.

Object Modelling

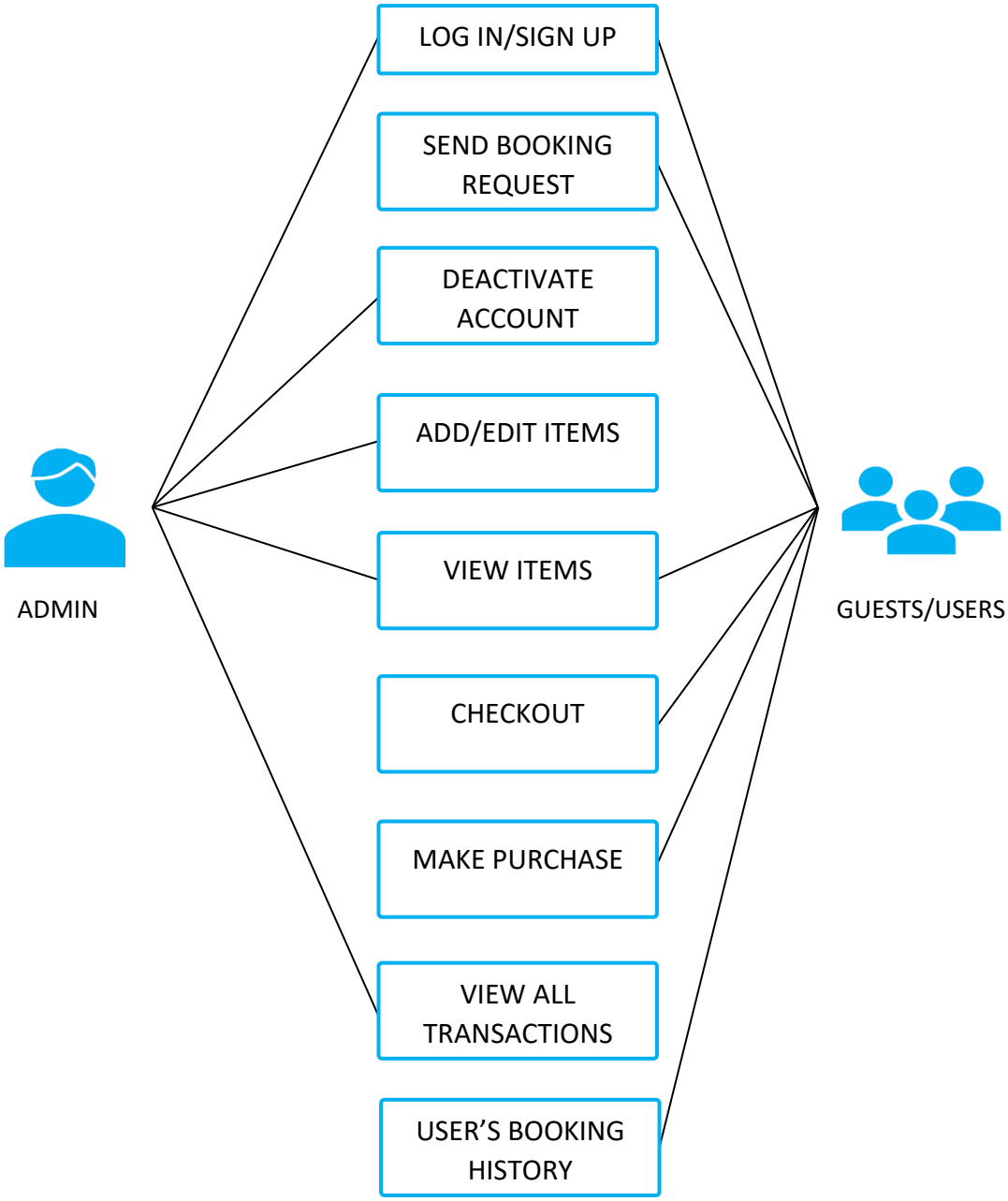

Figure 10. Use Case Diagram

The figure above summarizes the details of using the online reservation and their interactions with the system. The system administrator can log in to the system and delete account/s. The administrator can also view, add, and edit items in the system. Another system actor is the user/guest/client, who can create and sign in to the system using different interfaces. They can view items, make purchases, view their transactions, and check out those.

Class Diagram

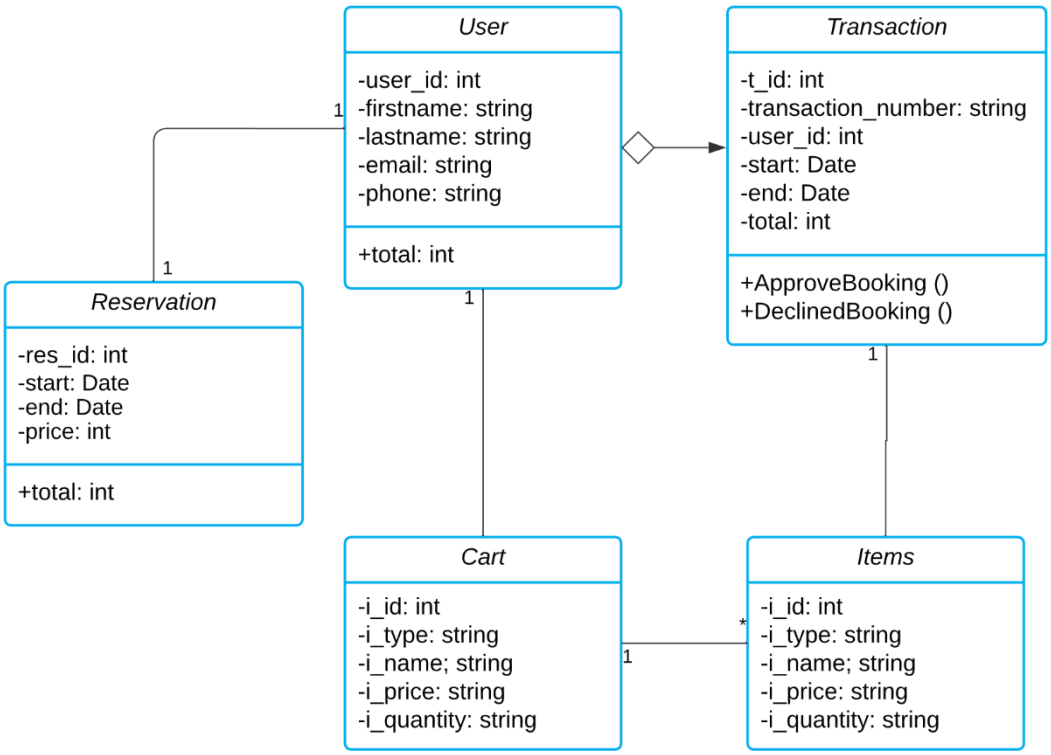

Figure 11. Class Diagram

The figure above presents a comprehensive view of the system's structure, documenting its constituent classes, including their attributes and methods. It also reveals a complex web of connections between elements, offering a detailed insight into the system's intricate design and interactions, which will surely engage and challenge our technical colleagues.

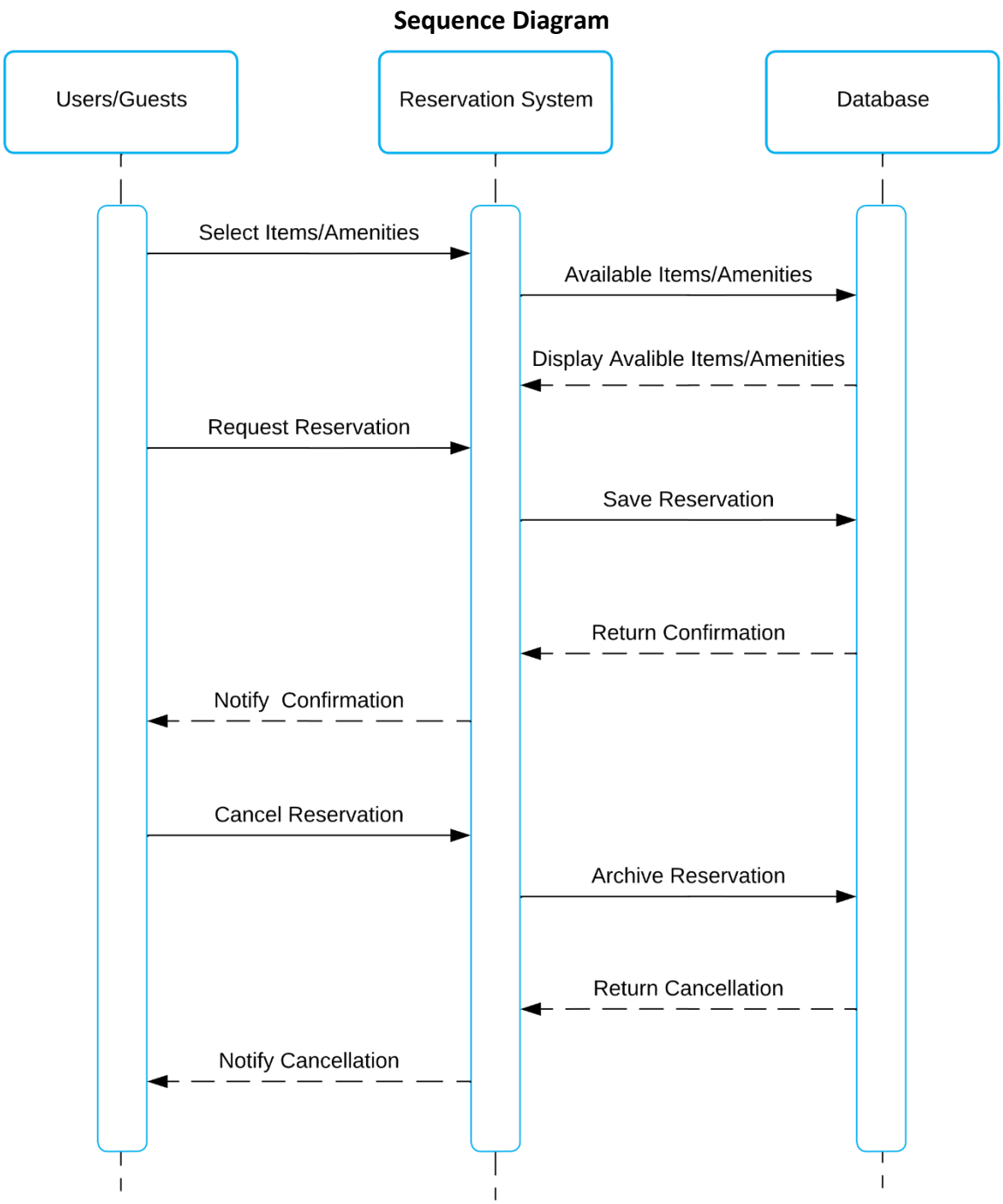

Figure 12. Sequence Diagram

This figure is a visual representation of the dynamic interactions between objects in a specific scenario or process. It vividly demonstrates the sequence of messages exchanged among objects over time, highlighting their collaborative efforts in achieving a specific task or goal. This emphasizes the crucial role each object plays in the system, making our software developers and system architects feel valued and important.

Design

Output and User-Interface Design

The Cugon Bamboo Resort's online reservation system incorporates a user-interface design that contains multiple forms to obtain important information from users and streamline the booking procedure. Each form is made with ease of use and simple navigation in mind to improve the user experience.

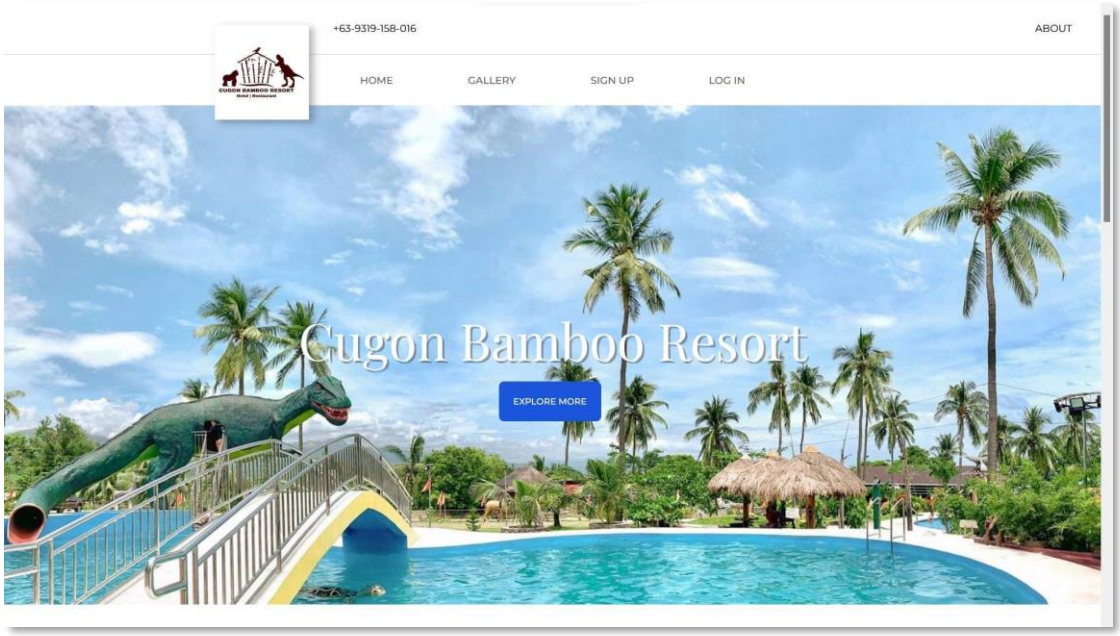

Figure 13. Homepage of ORS for Cugon Bamboo Resort

This figure showcases the user-friendly Online Reservation System (ORS) homepage for Cugon Bamboo Resort. The top navigation bar offers easy access to different sections of the website with options like 'Home,' 'Gallery,' 'Sign Up,' 'Login,' and 'About.' A contact number is also provided for direct inquiries. The central call-to-action button labeled 'Explore More' encourages users to delve into further details about the resort. This homepage is designed to instill confidence in potential guests by presenting the resort's picturesque setting and providing straightforward navigation options.

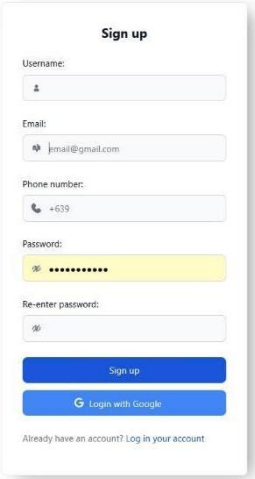

Figure 14. Sign Up Form

The figure above showcases a user-friendly 'Sign Up' form interface for the Cugon Bamboo Resort's online reservation system. This interface streamlines the account creation process, requiring users to input their username, email, phone number, and password. By making it easy to create a new account, we aim to encourage more users to register and enjoy the benefits of our online reservation system.

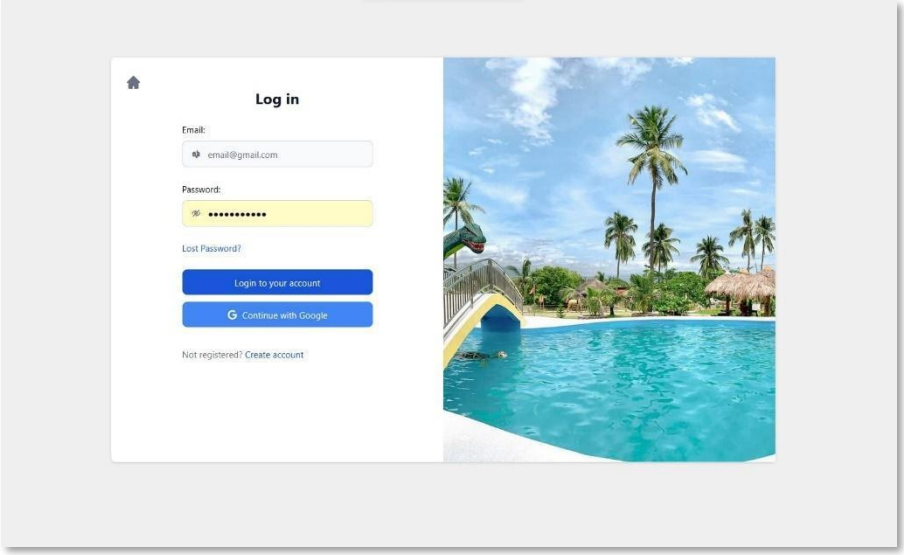

Figure 15. Login Form

The figure above showcases a user-friendly 'Log In' interface for Users or Customers and the Administrator of the ORS for Cugon Bamboo Resort. The left side of the screen features a convenient login form, where users can easily enter their email and password to access their accounts. Additional options include a 'Lost Password?' link for password recovery, a 'Login to your account' button, and an option to 'Continue with Google' for Google account login. A link at the bottom allows new users to create an account if they still need to register.

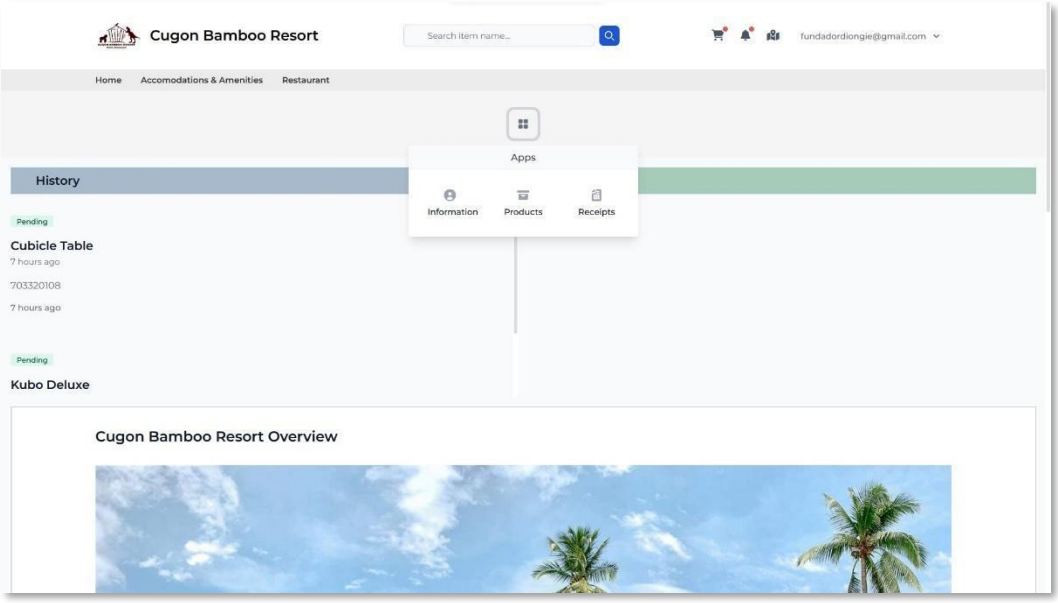

Figure 16. User Dashboard

This figure presents the comprehensive user dashboard for the Cugon Bamboo Resort's online reservation system. It provides a detailed overview of the user's booking history, including recent reservations such as a 'Cubicle Table' and 'Kubo Deluxe.' The dashboard features a user-friendly navigation menu at the top with options for 'Home,' 'Accommodations & Amenities,' and 'Restaurant.' Additionally, there is an 'Apps' dropdown menu offering quick access to 'Information,' 'Products,' and 'Receipts.' This intuitive interface is designed to empower guests, allowing them to easily manage their bookings and view resort information at their convenience.

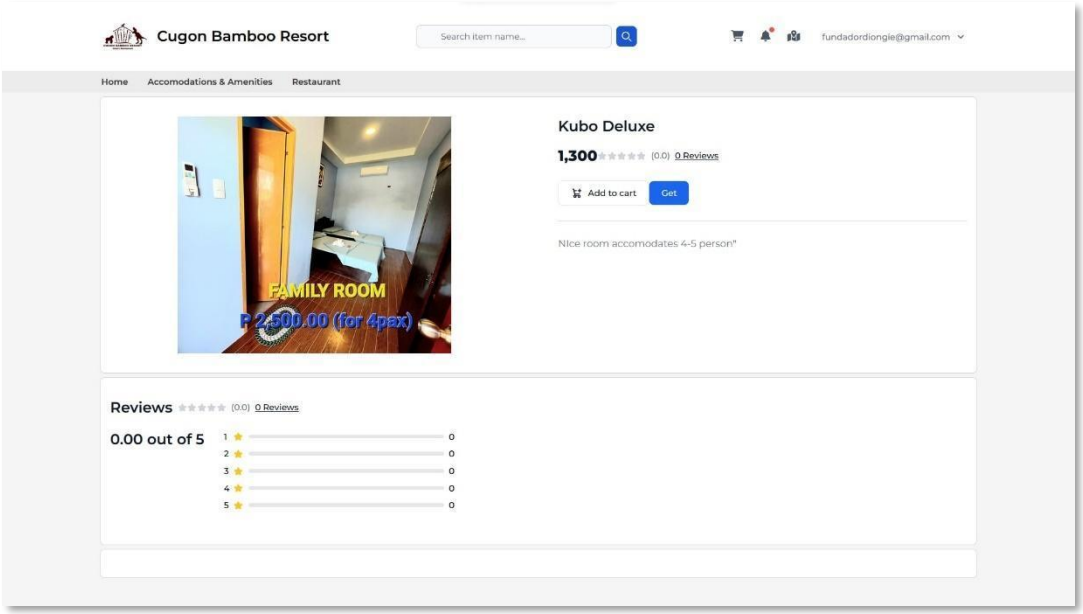

Figure 17. Room Information

This figure presents a comprehensive view of the room information available on the ORS for Cugon Bamboo Resort. It includes detailed descriptions of room types, prices, and accommodation capacities. You'll also find ratings, reviews, and images of the rooms, providing you with all the necessary details to make an informed decision about your lodging options.

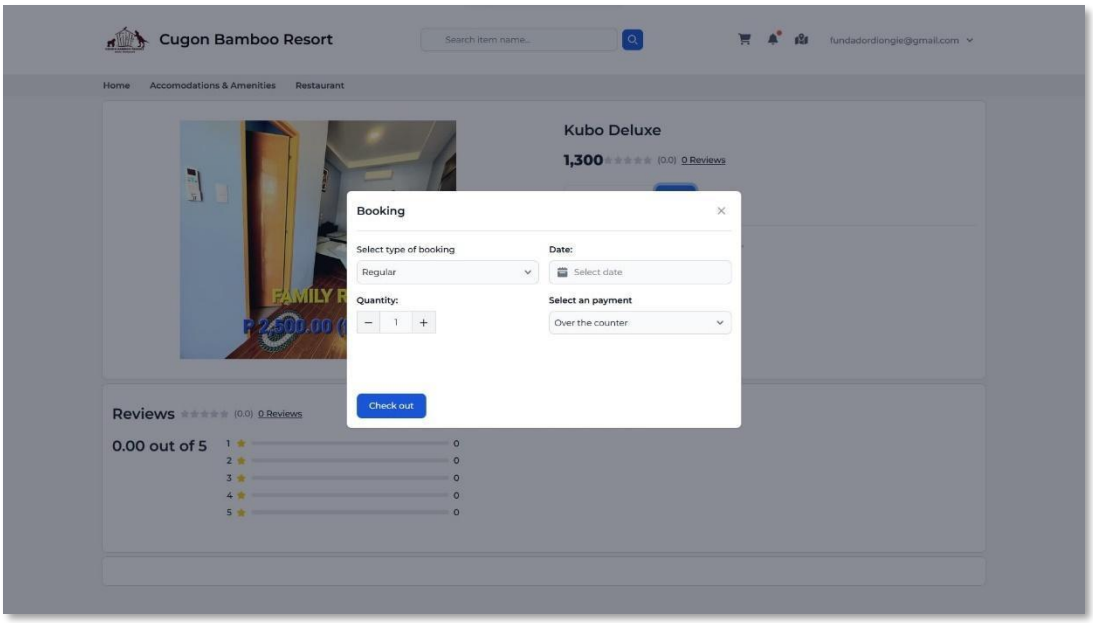

Figure 18. Booking Form

This figure showcases the user-friendly ORS booking form for Cugon Bamboo Resort. It streamlines the room reservation process, offering a straightforward selection of dates, quantity, and payment method, ensuring a hassle-free booking experience for our potential guests.

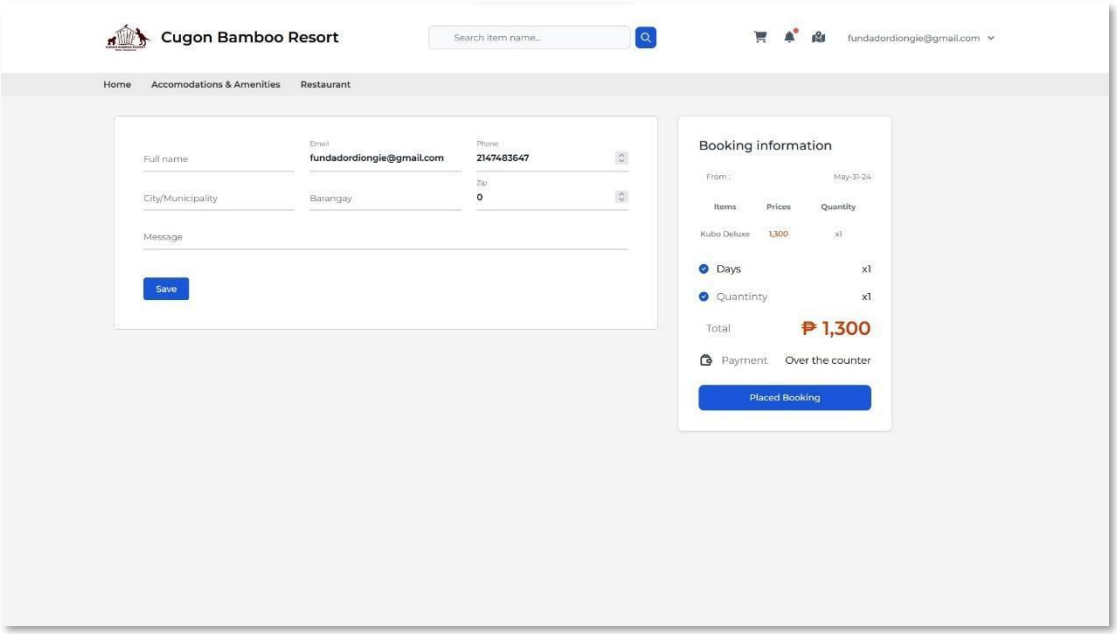

Figure 19. Booking & Payment Information

This figure illustrates the secure booking and payment information interface for Cugon Bamboo Resort. It features a form where users can confidently input their full name, email, phone number, city/municipality, barangay, and message, knowing their data is protected.

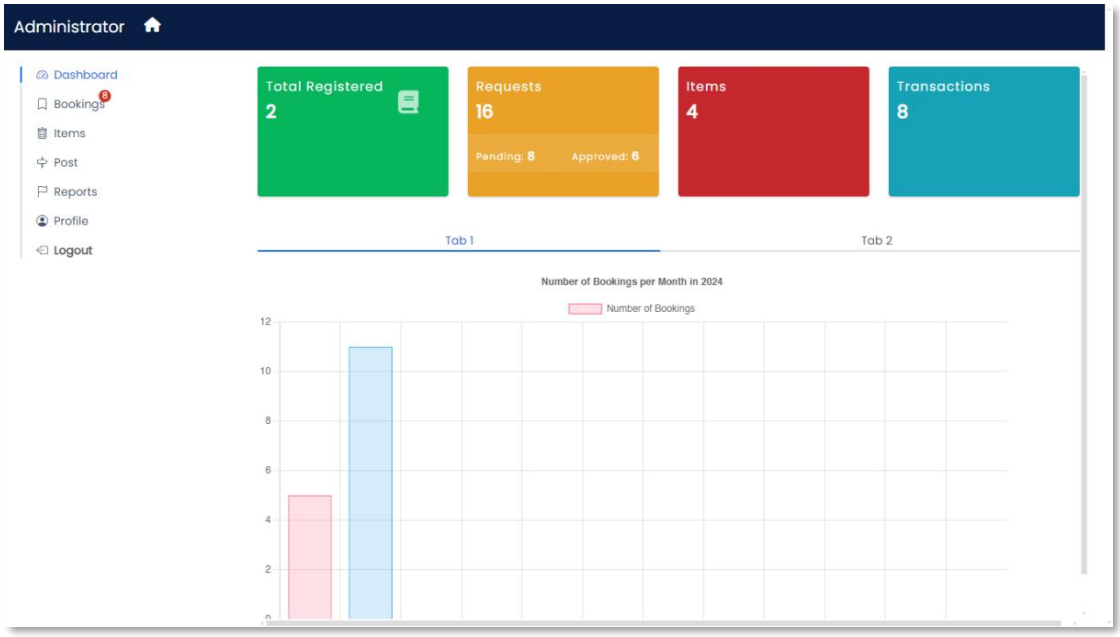

Figure 20. Admin Dashboard of ORS for Cugon Bamboo Resort

The figure presents the user-friendly ORS admin dashboard for Cugon Bamboo Resort. It features a robust navigation menu on the left, providing a plethora of options for Dashboard, Bookings, Items, Post, Reports, Profile, and Logout. This intuitive design empowers the staff and administrators, instilling confidence in their ability to efficiently manage the resort's operations.

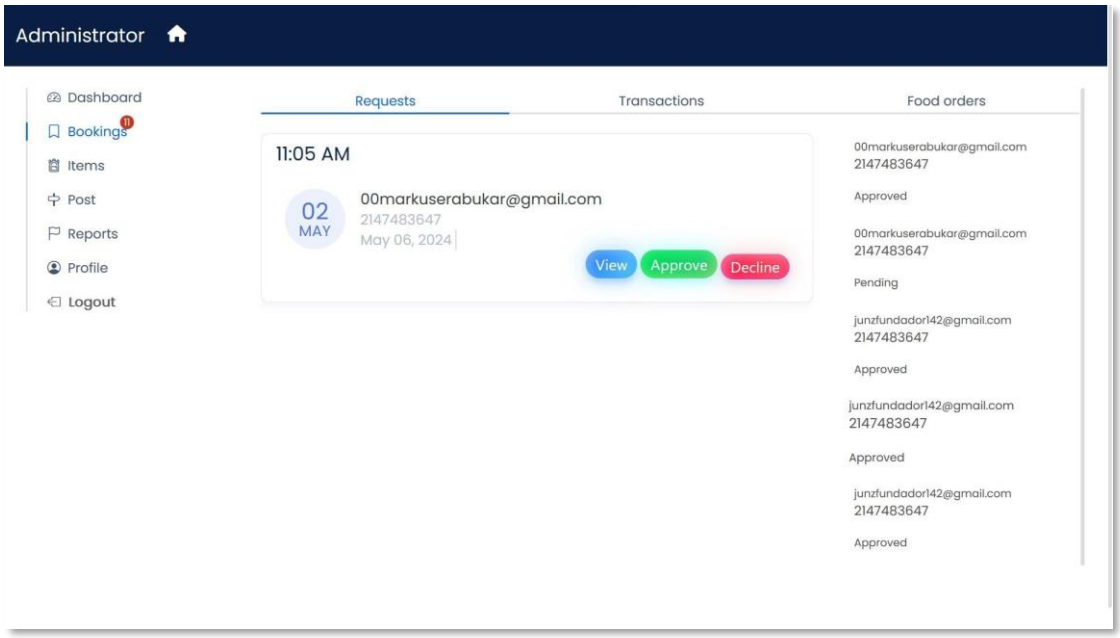

Figure 21. Bookings

The figure illustrates the efficient 'Bookings' section of the Cugon Bamboo Resort admin dashboard. It features a navigation menu with options like Dashboard, Bookings, Items, Post, Reports, Profile, and Logout. The main section is divided into three tabs: Requests (to manage booking requests), Transactions (to handle payment transactions), and Food Orders (to process food orders). The Requests tab displays a booking request with options to view, approve, or decline. The Transactions tab includes transaction details, and the Food Orders tab lists orders with their status as either Approved or Pending. This streamlined layout reassures the staff and administrators about their ability to handle these crucial tasks.

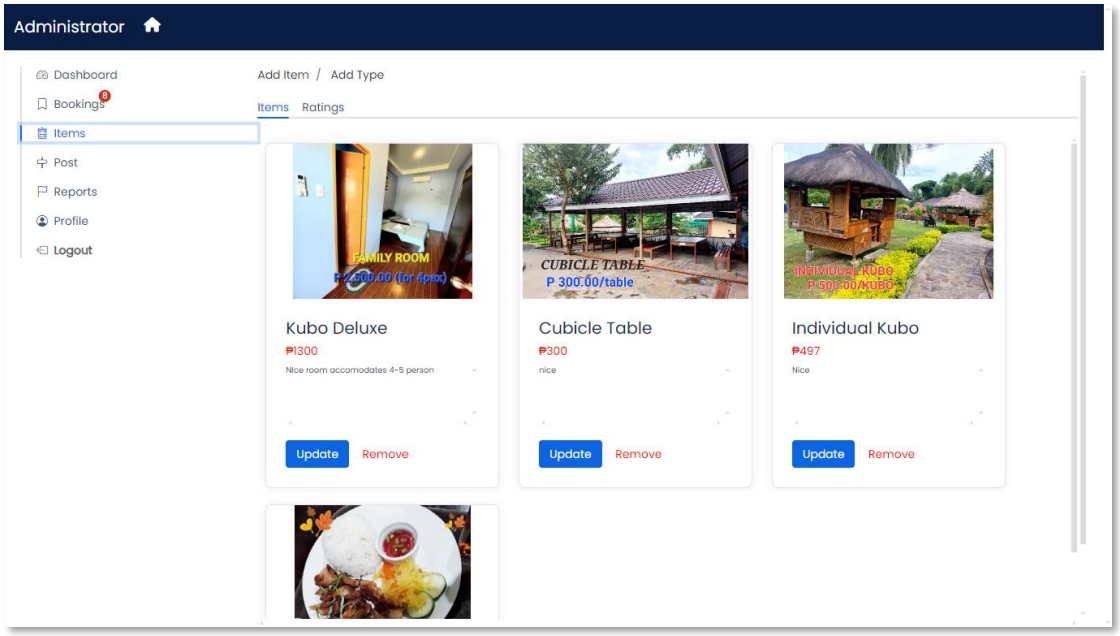

Figure 22. Items

This figure showcases the 'Bookings' section of the Cugon Bamboo Resort admin dashboard, which includes a navigation menu with options like Dashboard, Bookings, Items, Post, Reports, Profile, and Logout. The main section is divided into three tabs: Requests, Transactions, and Food Orders. The Items tab is a comprehensive list of accommodation options, complete with photos, names, prices, descriptions, and the 'Update' and 'Remove' buttons for each listing. This tab is a valuable resource for managing the resort's accommodation offerings.

Data Design

Entity Relationship Diagram

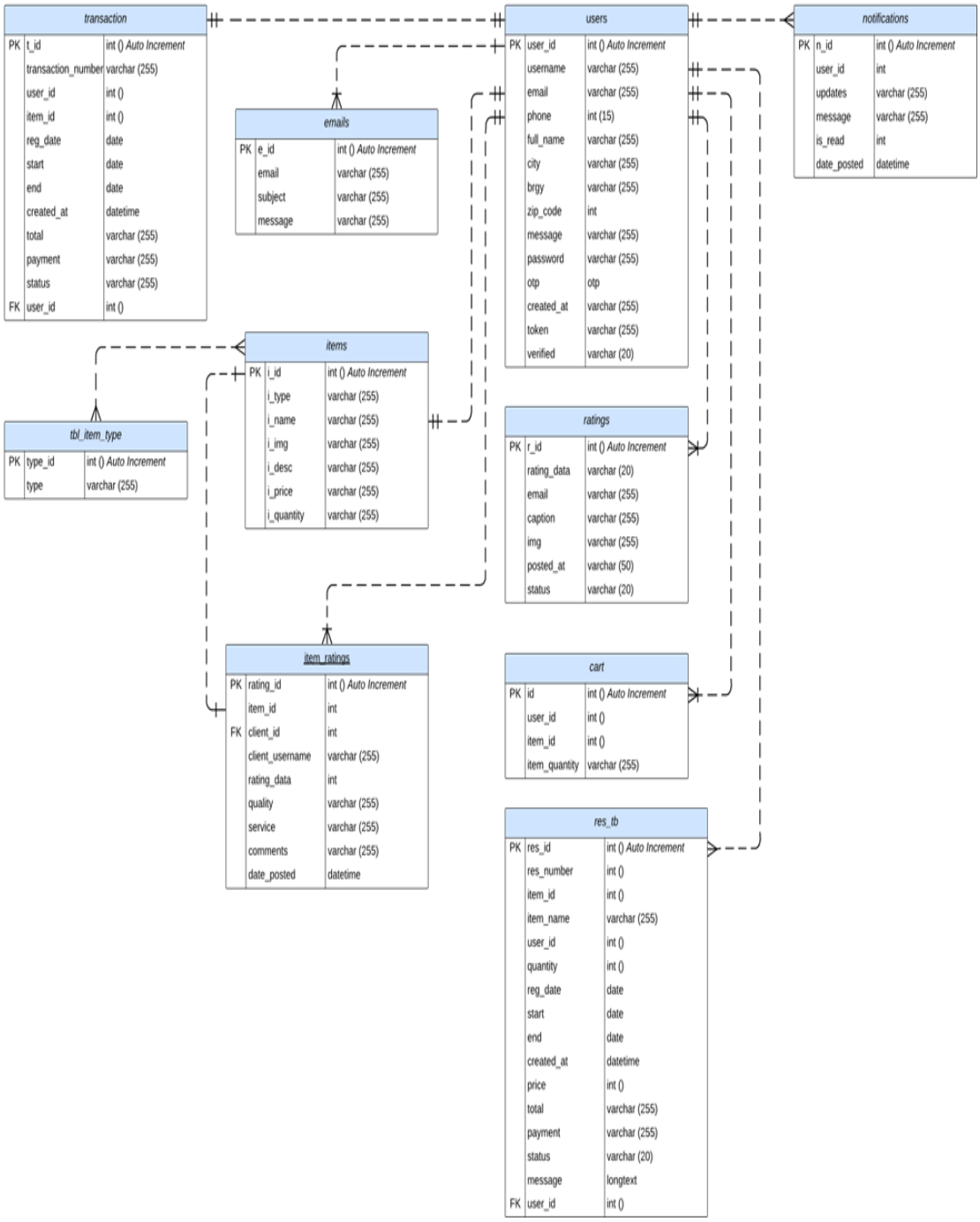

Figure 23. Entity Relationship Diagram

This figure shows the structure of the Cugon Bamboo Resort's online reservation system, including the entities, attributes, and relationships between them.

Data Dictionary

The Data Dictionary for Cugon Bamboo Resort outlines various data elements' structure, types, and purposes within the resort's database system. This dictionary serves as a reference guide for understanding how data is organized, stored, and managed, ensuring consistency and clarity across the resort's operations.

Table 4

Data Dictionary – users

Column	Type	Comment
user_id	int () *Auto Increment*	Unique ID, auto-incremented.
username	varchar (255)	User's chosen username.
email	varchar (255)	User's email address.
phone	int ()	User's phone number, max 15 digits.
full_name	varchar (255)	Full name of the user.
city	varchar (2550	City where the user resides.
brgy	varchar (255)	Barangay of the user.
zip_code	int ()	Zip code of the user's address.
message	varchar (255)	Message or additional notes.
password	varchar (255)	User's password.
otp	int ()	One-time password for verification.
created_at	varchar (255)	Date and time when the record was created.
token	varchar (255)	Authentication or session token.
verified	varchar (20)	Verification status (e.g., verified, pending).

This table shows the data dictionary for the "users" table within the Online Reservation System for Cugon Bamboo Resort. It contains essential fields like unique identifier, username, email, phone number, full name, location details, security features like password, one-time password, and authentication token, and records account creation and verification status for secure and efficient user management.

Table 5

Data Dictionary – rest_tb

Column	Type	Comment
res_id	int () *Auto Increment*	Unique ID, auto-incremented.
res_number	int ()	Reservation number or code.
item_id	int ()	Identifier for the reserved item.
item_name	varchar (255)	Name of item.
user_id	int ()	User's unique identifier linked to the reservation.
quantity	Int ()	Number of items reserved.
reg_date	Date	Date when the reservation was registered.
start	Date	Start date or time of the reservation.
end	Date	End date or time of the reservation.
created_at	Datetime	Date and time when the reservation was created.
price	int	Price per item or total price.
total	varchar (255)	Total price for the reservation.
payment	varchar (255)	Payment method or details.
status	varchar (20)	Current status of the reservation (e.g., confirmed, pending).
message	longtext	Additional message or notes related to the reservation.

This table presents the data dictionary for the 'reservations' table within the Online Reservation System for Cugon Bamboo Resort. It encompasses essential fields such as a unique identifier (res_id), reservation number (res_number), item details (item_id, item_name), user ID (user_id), and reservation specifics like quantity, dates, prices, payment method, status, and additional messages. This comprehensive structure ensures robust reservation management, providing reassurance about the system's capabilities.

Table 6

Data Dictionary – cart

Column	Type	Comment
id	int () *Auto Increment*	Unique ID, auto-incremented.
user_id	int	User's ID.
item_id	int	Item's ID.
item_quantity	varchar (255)	Quantity of the item.

This table shows the data dictionary for the "cart" table within the Cugon Bamboo Resort Online Reservation System. It includes essential fields such as a unique identifier for each cart entry, the user's ID, the item's ID, and the quantity of the item, enabling efficient tracking and management of items reserved by users.

Table 7

Data Dictionary – items

Column	Type	Comment
i_id	int () *Auto Increment*	Unique ID, auto-incremented.
i_type	varchar (255)	Type or category of the item.
i_name	varchar (255)	Name or title of the item.
i_img	varchar (255)	URL or path to the item's image.
i_desc	varchar (255)	Description of the item.
i_price	varchar (255)	Price of the item.
i_quantity	varchar (255)	Quantity or stock level of the item.

This table presents the data dictionary for the "items" table within the Cugon Bamboo Resort Online Reservation System. It encompasses important fields such as unique identifier (i_id), item type (i_type), name (i_name), image URL (i_img), description (i_desc), price (i_price), and quantity (i_quantity). These attributes provide comprehensive details for efficient management and representation of items available for reservation within the system.

Table 8

Data Dictionary – transactions

Column	Type	Comment
t_id	int () *Auto Increment*	Unique ID, auto-incremented.
transaction_number	Int ()	Transaction number or code.
user_id	Int ()	User's ID or identifier.
reg_date	date	Registration or transaction date.
start	date	Start date or time of the transaction.
end	date	End date or time of the transaction.
created_at	datetime	Date and time when the transaction was created.
total	varchar (255)	Total amount or price of the transaction.
payment	varchar (255)	Payment method or details.
status	varchar (255)	Current status of the transaction.

This table shows the data dictionary for the "transactions" table within the Online Reservation System for Cugon Bamboo Resort. It includes essential fields such as a unique identifier (t_id), transaction number (transaction_number), user ID (user_id), registration date (reg_date), start date (start), end date (end), creation timestamp (created_at), total amount (total), payment method (payment), and transaction status (status). These attributes facilitate comprehensive management and tracking of transactions, ensuring efficient handling of reservation activities at Cugon Bamboo Resort.

Table 9

Data Dictionary – ratings

Column	Type	Comment
r_id	int () *Auto Increment*	Unique ID, auto-incremented.
rating_data	varchar (20)	Rating value or data (e.g., 4.5 stars).
email	varchar (255)	User's email address.
caption	varchar (255)	Caption or description for the rating.
img	varchar (255)	URL or path to the image associated with the rating.
status	varchar (20)	Status of the rating (e.g., approved, pending).

This table shows the data dictionary for the "ratings" table within the Online Reservation System for Cugon Bamboo Resort. It encompasses essential fields such as a unique identifier (r_id), rating data (rating_data), user's email address (email), caption (caption), image URL (image), and status (status) of the rating, ensuring comprehensive management and tracking of user feedback.

Table 10

Data Dictionary – item_ratings

Column	Type	Comment
rating_id	int () *Auto Increment*	Unique ID, auto-incremented.
item_id	int ()	ID of the item being rated.
client_id	int ()	ID of the client who provided the rating.
client_username	varchar (255)	Username of the client.
rating_data	int ()	Numerical rating value.
quality	varchar (255)	Comments on the quality of the item.
service	varchar (255)	Comments on the service provided.
comments	varchar (255)	Additional comments from the client.
date_posted	datetime	Date and time when the rating was posted.

This table illustrates the data dictionary for the "item_ratings" table within the Online Reservation System for Cugon Bamboo Resort. It comprises essential fields including the unique identifier (rating_id), item ID (item_id), client ID (client_id), client username (client_username), rating data (rating_data), quality comments (quality), service comments (service), additional comments (comments), and date posted (date_posted). These attributes facilitate thorough management and analysis of client feedback on specific items, enhancing service quality and customer satisfaction levels at Cugon Bamboo Resort.

Table 11

Data Dictionary – food_orders

Column	Type	Comment
order_id	int () *Auto Increment*	Unique ID, auto-incremented.
client_name	varchar (255)	Name of the client placing the order.
food	varchar (255)	Description or name of the food items ordered.
drinks	varchar (255)	Description or name of the drinks ordered.
total	int ()	Total cost of the order.
order_status	varchar (255)	Current status of the order (e.g., pending, completed).

This table presents the data dictionary for the "food_orders" table within the Online Reservation System for Cugon Bamboo Resort. It includes essential fields such as the unique identifier (order_id), client name (client_name) placing the order, description or name of food items ordered (food), description or name of drinks ordered (drinks), total cost of the order (total), and current status of the order (order_status). These attributes play a crucial role in our operations, enabling effective management and tracking of food and drink orders, which in turn ensures smooth service and customer satisfaction at Cugon Bamboo Resort.

Table 12

Data Dictionary – notifications

Column	Type	Comment
n_id	int () *Auto Increment*	Unique ID, auto-incremented.
user_id	int ()	ID of the user receiving the notification.
updates	varchar (255)	Summary or type of update being notified.
message	varchar (255)	Notification message content.
is_read	int ()	Indicator if the notification has been read
date_posted	datetime	Date and time when the notification was posted.

This table presents the data dictionary for the "notifications" table within the Online Reservation System for Cugon Bamboo Resort. It includes essential fields such as the unique identifier (n_id), user ID (user_id) of the recipient, summary or type of update being notified (updates), notification message content (message), indicator if the notification has been read (is_read), and date and time when the notification was posted (date_posted). This structure, with its efficient design, facilitates effective management and delivery of messages, ensuring timely communication and user engagement within the reservation system at Cugon Bamboo Resort.

Table 13

Data Dictionary – payment_method

Column	Type	Comment
id	int () *Auto Increment*	Unique ID, auto-incremented.
payment	varchar (255)	Name or type of the payment method.

This table presents the data dictionary for the "payment_method" table within the Cugon Bamboo Resort Online Reservation System. It encompasses essential fields, including the unique identifier (id) for each payment method and the name or type of the payment method (payment). This structured format facilitates efficient management and selection of payment methods, contributing to a seamless reservation experience at Cugon Bamboo Resort.

Table 14

Data Dictionary – resto_products

Column	Type	Comment
id	int () *Auto Increment*	Unique ID, auto-incremented.
img	varchar (255)	URL or path to the product's image.
name	varchar (255)	Name of the product.
description	varchar (255)	Description of the product.
price	varchar (255)	Price of the product.

This table illustrates the data dictionary for the "resto_products" table within the Online Reservation System for Cugon Bamboo Resort. It encompasses crucial fields including the unique identifier (id), URL or path to the product's image (img), product name (name), description (description), and price (price) of the product. This structured format ensures effective management and presentation of restaurant products, enhancing the dining experience for guests at Cugon Bamboo Resort.

Development

Table 15

Software Specification

Software	Use
Visual Studio Code	Code editor for development.
WampServer	Local server for web development.
Google Chrome/Mozilla Firefox	Web browsers for testing and browsing.
MySQL	Database management system.
HTML	Markup language for web pages.
CSS	Styling language for web pages.
JavaScript	JavaScript library for building user interfaces.
PHP	Server-side scripting language.

The software specification for the Online Reservation System for Cugon Bamboo Resort is a comprehensive set of tools, each designed to complement the other. It includes Visual Studio Code for development, WampServer as the local server, and Google Chrome/Mozilla Firefox for testing. MySQL is the database management system, while HTML structures web pages and CSS styles them. JavaScript enhances user interfaces, and PHP handles server-side scripting. These tools, working in perfect harmony with the recommended hardware, support the system's functionality within the resort.

Table 16

Hardware Specification

Hardware	Specification
RAM	8GB and above
HARD DISK	512GB SSD
PROCESSOR	Intel Core i7 and above
OPERATING SYSTEM	Windows 10 and above
DISPLAY	24-inch Full HD Monitor

The recommended hardware specifications for running the Online Reservation System for Cugon Bamboo Resort are designed to deliver high performance and ensure a seamless user experience. They include 8GB RAM or above, a 512GB SSD for storage, an Intel Core i7 processor or higher, Windows 10 or above as the operating system, and a 24-inch Full HD monitor. These specifications, carefully chosen, guarantee smooth operation and an optimal user experience.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATION

This chapter provides an extensive overview of the study's findings, conclusions, and recommendations for future improvements. It combines the essential lessons from creating and deploying an Online Reservation System for Cugon Bamboo Resort, highlighting accomplishments and improvement areas.

Summary of Findings

The study focuses on developing an online reservation system for Cugon Bamboo Resort using HTML, CSS, JavaScript, and PHP/MySQL. This system aims to enhance consumer convenience by streamlining the reservation process and introducing a GCash payment option, offering specific benefits like faster transactions, more user-friendly interactions and lower fees. This initiative is expected to reduce operating expenses and improve efficiency. While technology can enhance user experience and operational productivity, the research identified limitations in payment methods as the system exclusively accepts GCash, potentially limiting user flexibility. Additionally, the study encountered constraints due to a small sample size and inconsistent data quality, which may affect the generalizability and reliability of the findings.

Conclusions

The implementation of an online reservation system for Cugon Bamboo Resort has yielded numerous significant benefits, including enhanced user convenience, cost savings, and operational efficiency. Leveraging modern web technologies has facilitated the creation of a user-friendly design and robust functionality, resulting in a positive user experience. However, limited payment choices indicate a vital area for future development, such as integrating PayPal or credit card payments, to offer users more flexibility. The constraints related to sample size and data quality emphasize the necessity for further research with a larger and more diverse dataset to validate our findings comprehensively. Despite these challenges, the study underscores the

potential of technology integration in reservation systems to improve customer satisfaction and streamline corporate operations.

Recommendations

The newly-created online reservation system is highly recommended to the following:

Cugon Bamboo Resort Management

The management of Cugon Bamboo Resort can take pride in the successful implementation of the online reservation system. This system has not only optimized our reservation procedure but also significantly improved customer satisfaction and operational efficiencies. We recommend further enhancing the system by incorporating additional payment options like credit cards and PayPal to offer our customers even more flexibility and convenience.

Other Resorts and Hotels

For other resorts and hotels, the adoption of a similar online reservation system can be a game-changer. It simplifies operations, saves money, and provides customers with a more straightforward booking experience. This system has the potential to revolutionize the hospitality industry, inspiring others to follow suit.

Future Researchers

The study's findings provide an excellent basis for future researchers who want to research further improvements to the online reservation system. More research with bigger sample sizes and diverse data sets will be required to confirm and enhance these findings.

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APPENDICES

Appendix A. Project Team Assignments Form

TEAM ALIAS	**SCIENCE & FAITH**		
NAME	**SIGNATURE**	**POSITION**	**CONTACT #**
DIONGIE C. FUNDADOR		PROJECT MANAGER / PROGRAMMER	09319158016
GINO Q. MALCO		UI DESIGNER / TECHNICAL WRITER	09289567523
KERWIN P. BABOR		TECHNICAL WRITER	09945954769
MARY JOYCE F. CANDIDO		TECHNICAL WRITER	09975240254
MYCA C. CAPINGCOT		TECHNICAL WRITER	09605111037

Appendix B. ***Pre-Proposal Statement Template***

COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

CSC 402 / ITS 403 – Capstone Project 2

PROJECT TITLE:	ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT
PROPONENTS/RESEARCHERS:	1. DIONGIE C. FUNDADOR 2. GINO Q. MALCO 3. KERWIN P. BABOR 4. MARY JOYCE F. CANDIDO 5. MYCA C. CAPINGCOT
SCOPE OF THE STUDY:	This study focuses on using HTML, CSS, JavaScript, and PHP/MySQL to develop a user-friendly and dynamic website for Cugon Bamboo Resort. The aim is to enhance customer convenience and streamline operations, leading to cost savings and improved efficiency. By implementing these technologies, the resort can optimize its processes and extend its reach to a broader audience through various online channels.
LIMITATION OF THE STUDY:	The study has limitations, including offering only GCash payments via email without support for API gateways or other options like PayPal, limiting payment flexibility. Additionally, issues with sample size and data quality may affect the evaluation of online reservation systems, potentially limiting the generalizability and reliability of findings. Accurate interpretation of results requires acknowledging these limitations.
PROJECT DESIGN/DEVELOPMENT PLAN	
HARDWARE SPECIFICATION	Processor: At least Intel Core i5 and Above HDD: 512 GB RAM: 8 GB and above
SOFTWARE SPECIFICATION	JavaScript, MySQL, HTML, CSS, PHP, WampServer, Google Chrome/Mozilla Firefox, Visual Studio Code

Appendix C. ***Project Working Title Form***

Project Working Title Form
Computer Science and Information Technology (CSIT) Department
ITS 403 – Capstone Project 2

Proponents/Researchers:

1.) DIONGIE C. FUNDADOR
2.) GINO Q. MALCO
3.) KERWIN P. BABOR
4.) MARY JOYCE F. CANDIDO
5.) MYCA C. CAPINGCOT

Proposed Project Title:

ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT

Submitted by:	Noted by:
DIONGIE C. FUNDADOR Project Manager Date: _____	**ZENAIDA D. CALUMPANG, MSIT, Ed.D.** Subject Instructor Date: _____
Recommending Approval: **MARK ANTHONY A. CATALUÑA** Panel Member, CSIT Department Date: _____ **JETTY A. CASIDO** Panel Member, CSIT Department Date: _____ **MICHAEL C. YLEAÑA** Panel Member, CSIT Department Date: _____	Approved: **ZENAIDA D. CALUMPANG, MSIT, Ed.D.** Dean, College of Arts and Sciences Date: _____

Appendix D. ***Invitation Letter to the Panel Member 1***

Negros Oriental State University
COLLEGE OF ARTS AND SCIENCES
Campus 1, Bais City

May 2, 2023

MR. MARK ANTHONY A. CATALUÑA
Faculty, CAS
Negros Oriental State University
Subject: **FINAL ORAL DEFENSE FOR CAPSTONE PROJECT**

In compliance with the pertinent provisions of CMO No. 53, series of 2006, and in order to ensure that the quality of the Capstone Projects of the BS Information Technology Students are in accordance with the University's standards, it is imperative that the said projects be subjected to thorough evaluation and final defense.

In view of the foregoing, you are cordially invited to be the Chairperson of the Panel of Evaluators for the project entitled "**ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT**" by Diongie C. Fundador, et al.

Below is the information relative to the date and place of final defense.

Date: May 2, 2023
Time: 10:00-11:00 AM
Place: CAS AVR

Attached is the copy if the manuscript and Evaluation Sheet for your guidance.

Thank you for your cooperation and support.

Ma. ARCHEL LIMAN CALUNSAG
Project Adviser
Negros Oriental State University

Appendix E. ***Invitation Letter to the Panel Member 2***

Negros Oriental State University
COLLEGE OF ARTS AND SCIENCES
Campus 1, Bais City

May 2, 2023

MR. JETTY A. CASIDO
Faculty, CAS
Negros Oriental State University
Subject: **FINAL ORAL DEFENSE FOR CAPSTONE PROJECT**

In compliance with the pertinent provisions of CMO No. 53, series of 2006, and in order to ensure that the quality of the Capstone Projects of the BS Information Technology Students are in accordance with the University's standards, it is imperative that the said projects be subjected to thorough evaluation and final defense.

In view of the foregoing, you are cordially invited to be the Chairperson of the Panel of Evaluators for the project entitled "**ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT**" by Diongie C. Fundador, et al.

Below is the information relative to the date and place of final defense.

Date: May 2, 2023
Time: 10:00-11:00 AM
Place: CAS AVR

Attached is the copy if the manuscript and Evaluation Sheet for your guidance.

Thank you for your cooperation and support.

Ma. ARCHEL LIMAN CALUNSAG
Project Adviser
Negros Oriental State University

Appendix F. ***Invitation Letter to the Panel Member 3***

Negros Oriental State University
COLLEGE OF ARTS AND SCIENCES
Campus 1, Bais City

May 2, 2023

MR. MICHEAL C. YLEAÑA
Faculty, CAS
Negros Oriental State University
Subject: **FINAL ORAL DEFENSE FOR CAPSTONE PROJECT**

In compliance with the pertinent provisions of CMO No. 53, series of 2006, and in order to ensure that the quality of the Capstone Projects of the BS Information Technology Students are in accordance with the University's standards, it is imperative that the said projects be subjected to thorough evaluation and final defense.

In view of the foregoing, you are cordially invited to be the Chairperson of the Panel of Evaluators for the project entitled "**ONLINE RESERVATION SYSTEM FOR CUGON BAMBOO RESORT**" by Diongie C. Fundador, et al.

Below is the information relative to the date and place of final defense.

Date: May 2, 2023
Time: 10:00-11:00 AM
Place: CAS AVR

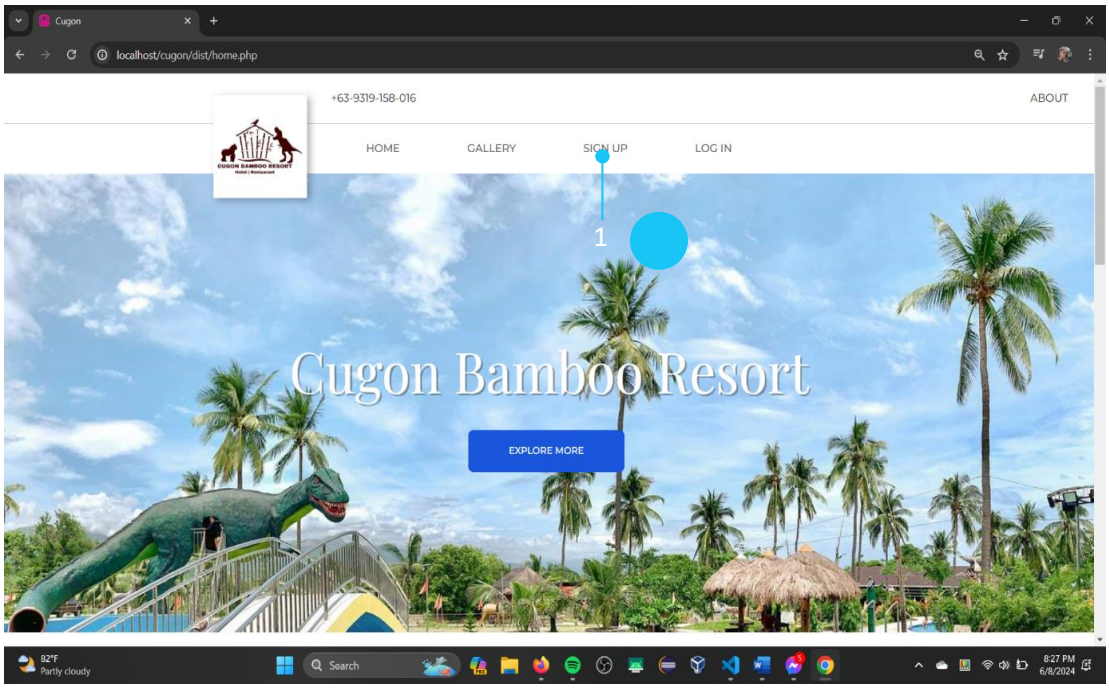
Attached is the copy if the manuscript and Evaluation Sheet for your guidance.

Thank you for your cooperation and support.

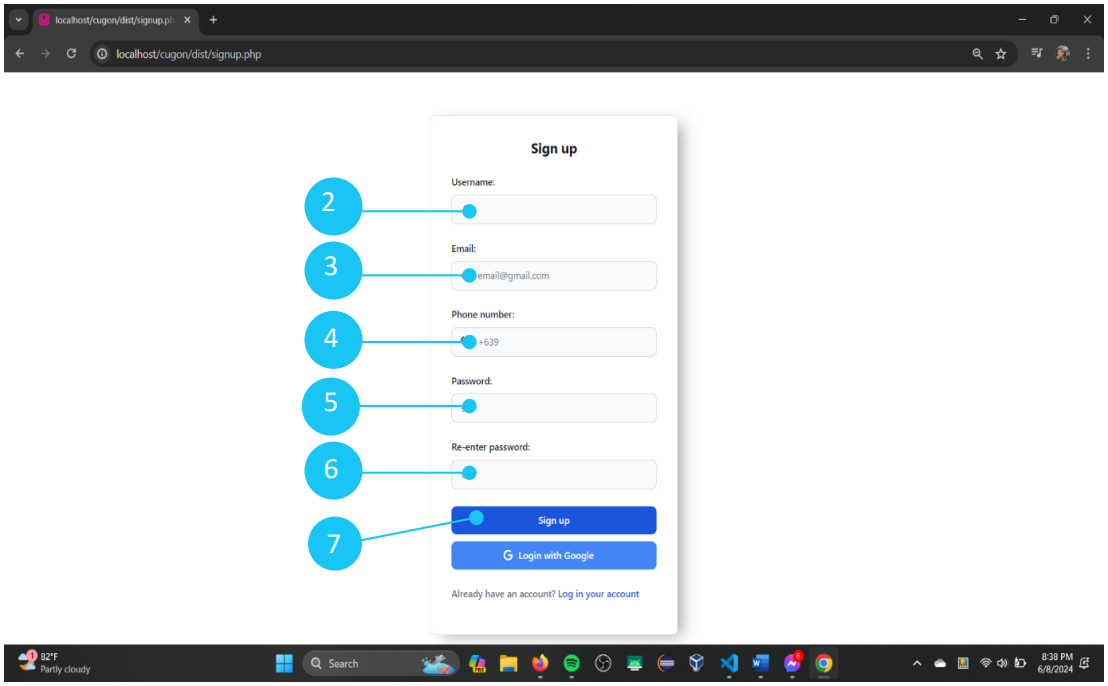
Ma. ARCHEL LIMAN CALUNSAG
Project Adviser
Negros Oriental State University

Appendix G. ***Location Map of Cugon Bamboo Resort***

Appendix H. *User's manual*



1. Click the ***SIGN-UP*** button to sign up



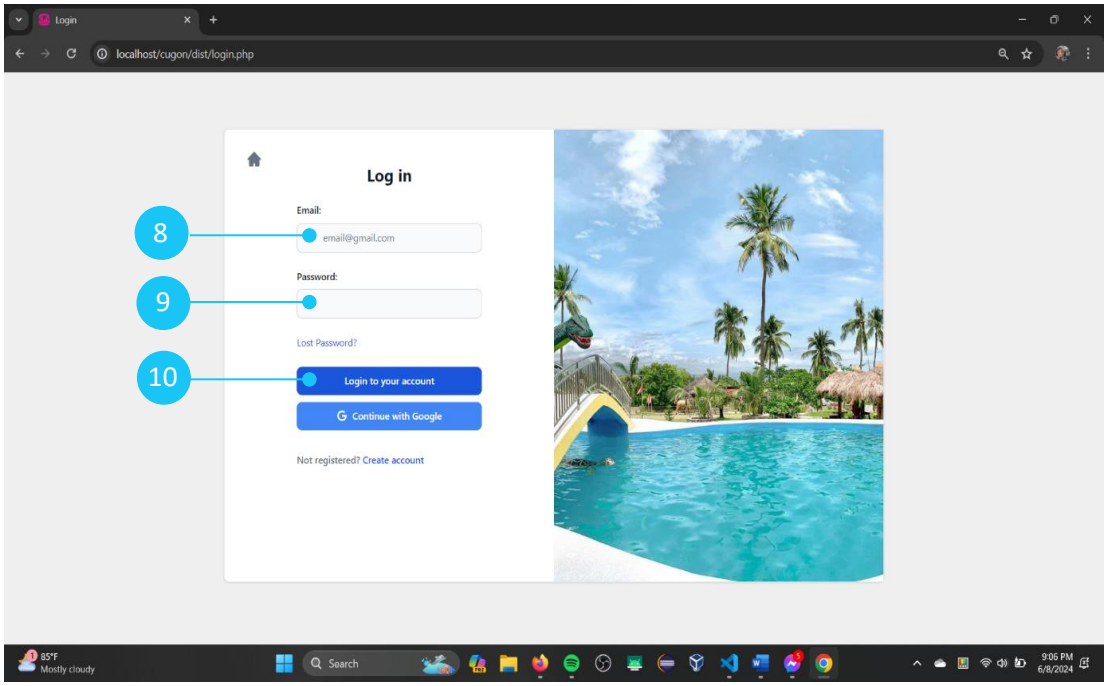
2. Input ***Username***

3. Input a valid ***email***

4. Input ***Phone number***

5. Input ***Password***

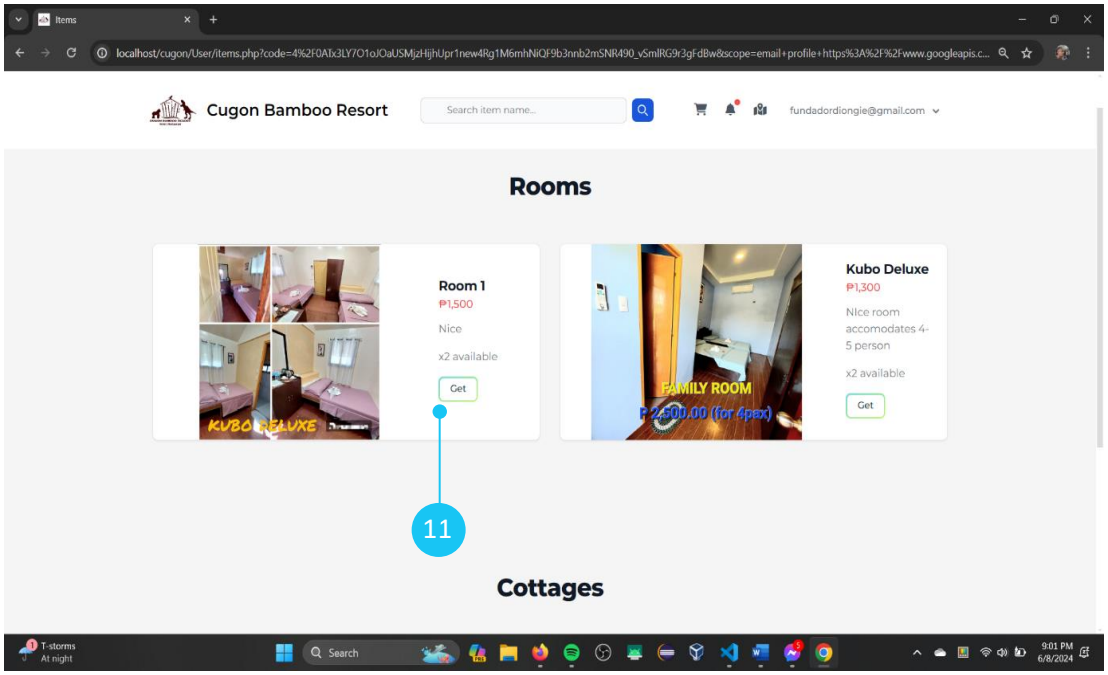
6. Re-type ***Password***



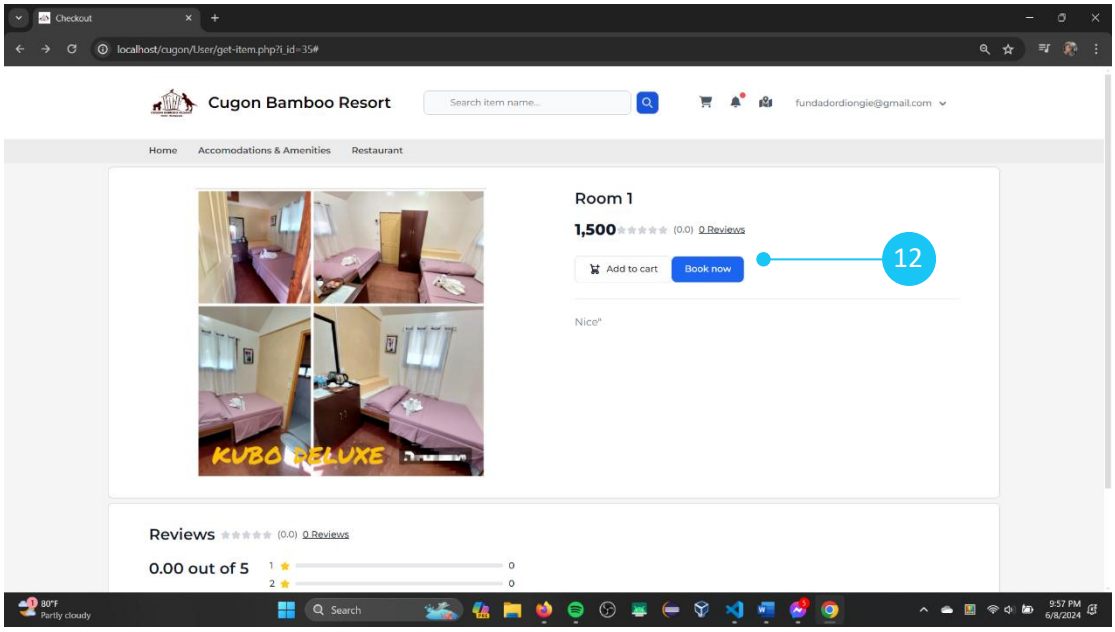
8. Input ***email***

9. Input ***password***

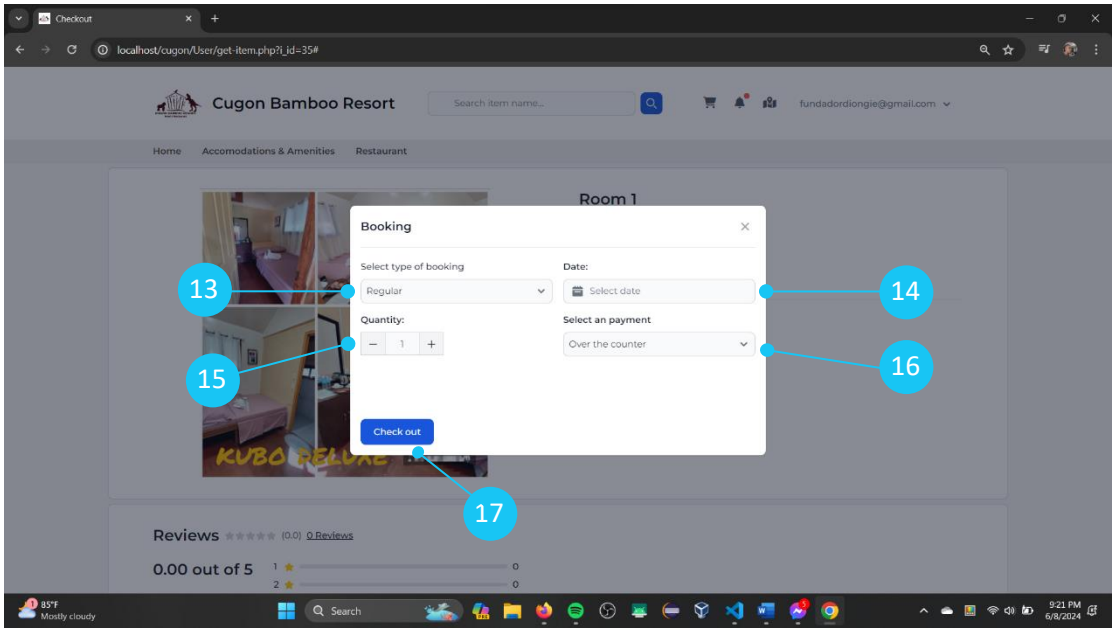
10. Click to continue ***Log in***



11. Click to ***Get*** for your prefer room or cottage



12. Click to ***Book now*** to continue booking



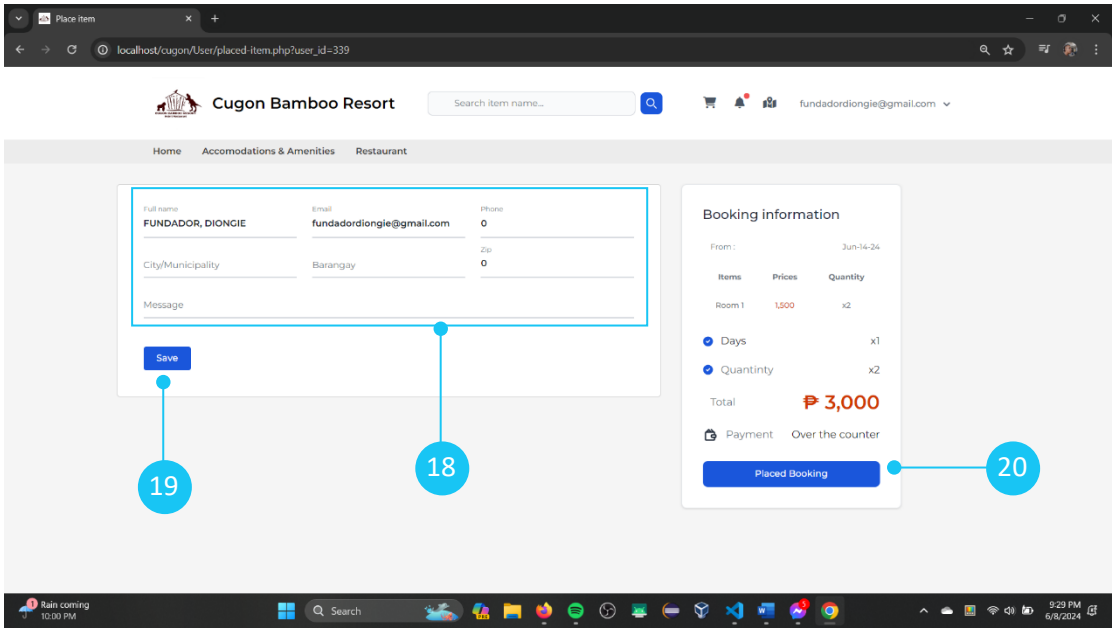
13. Select ***type of booking***

14. Select a valid ***date***

15. Select ***quantity***

16. Select ***type of payment***

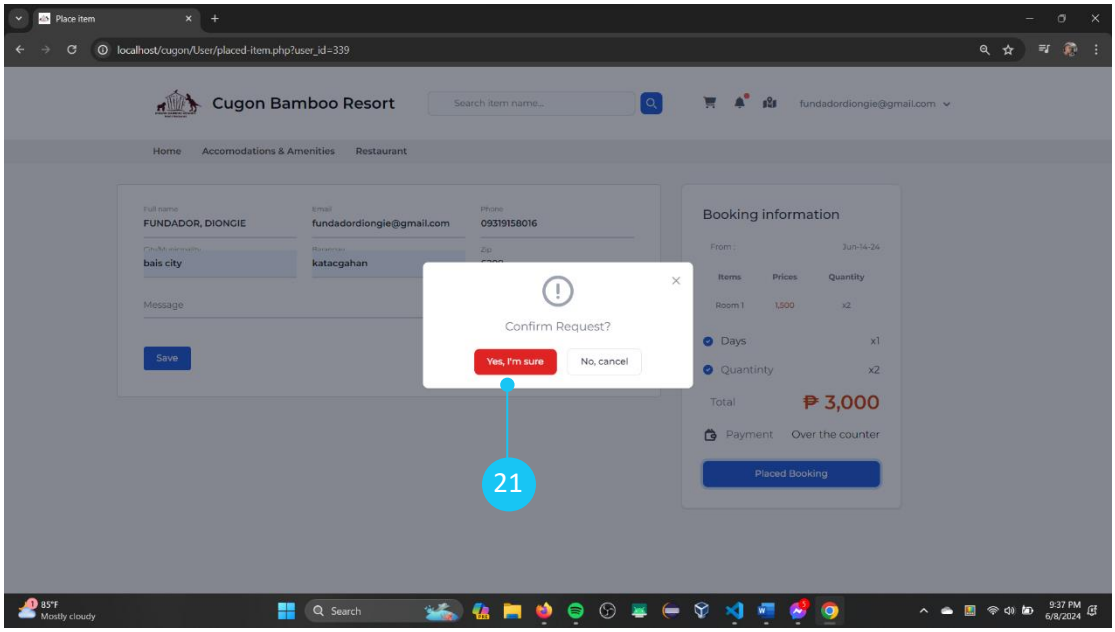
17. Click ***Check out*** to continue



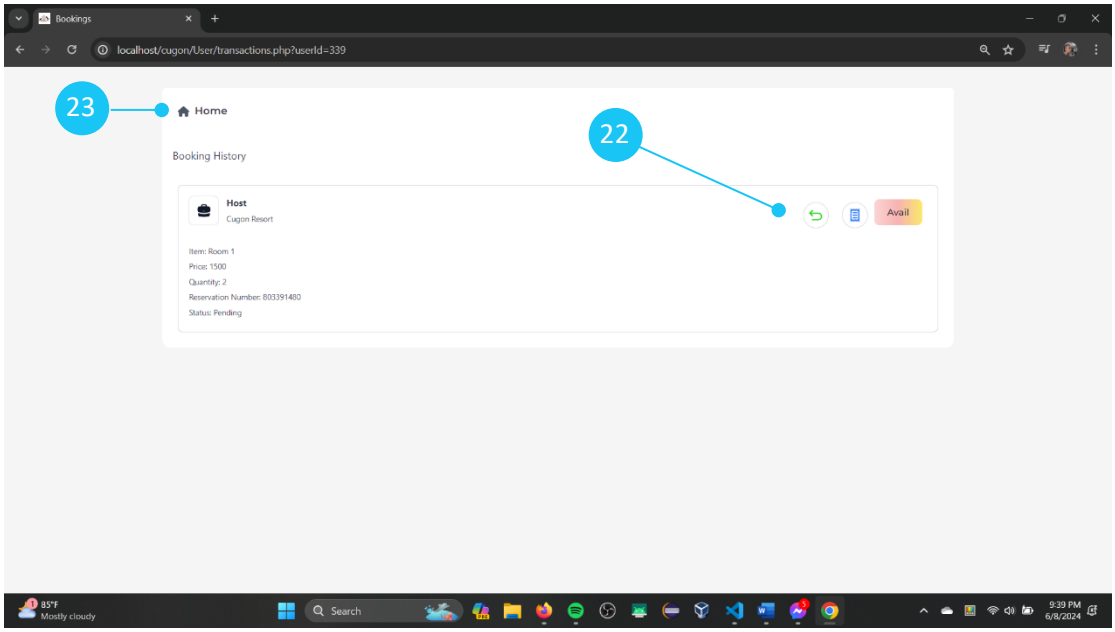
18. Input details

19. Click ***save***

20. Click ***Place Booking*** to continue

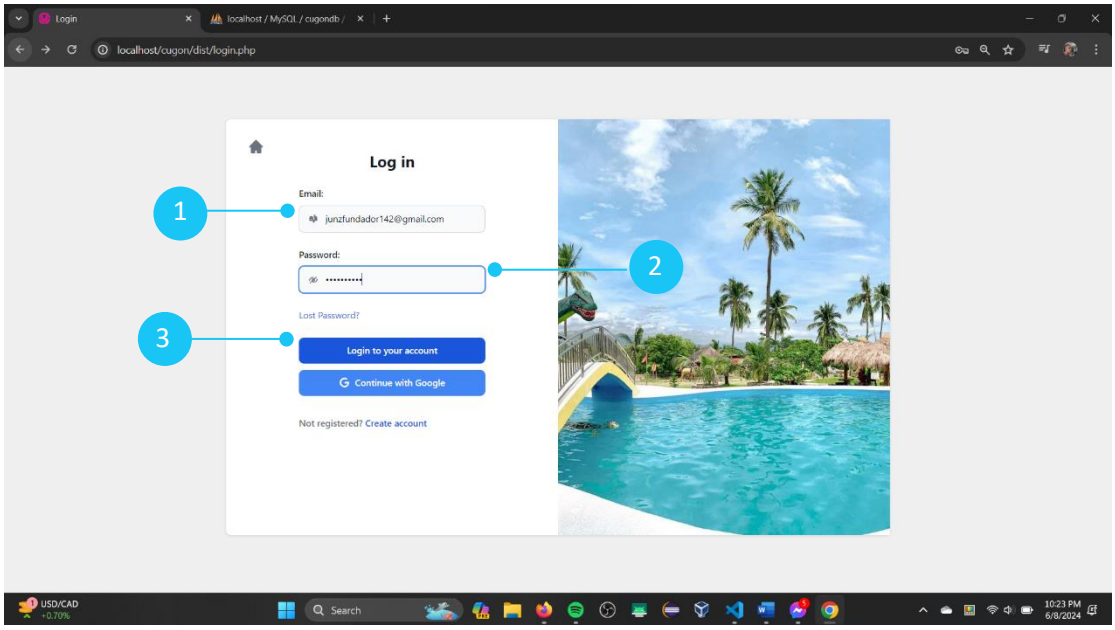


21. Click ***Yes, I'm sure*** to confirm request.



22. Click to Cancel, View Receipt and Avail the item again.

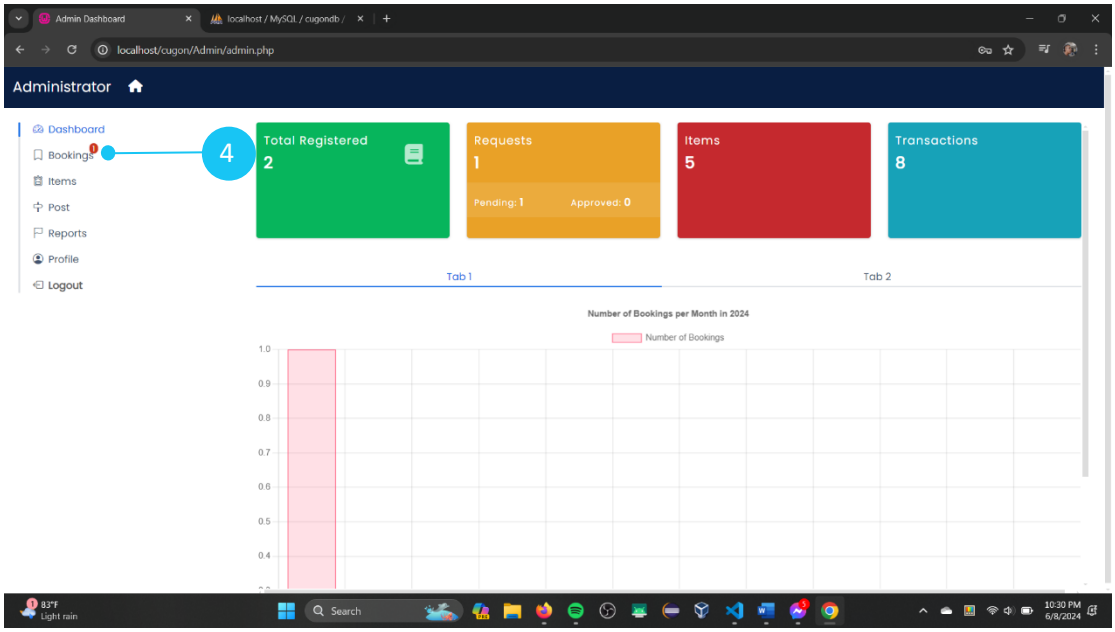
23. Click to go to ***Home***



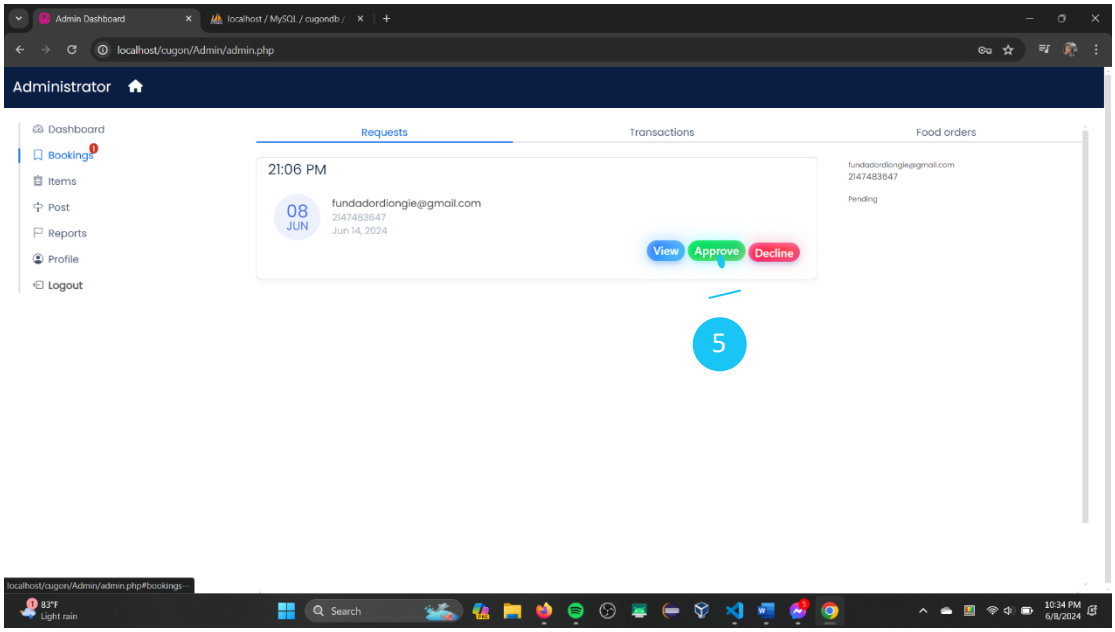
1. Input admin email.

2. Input admin password.

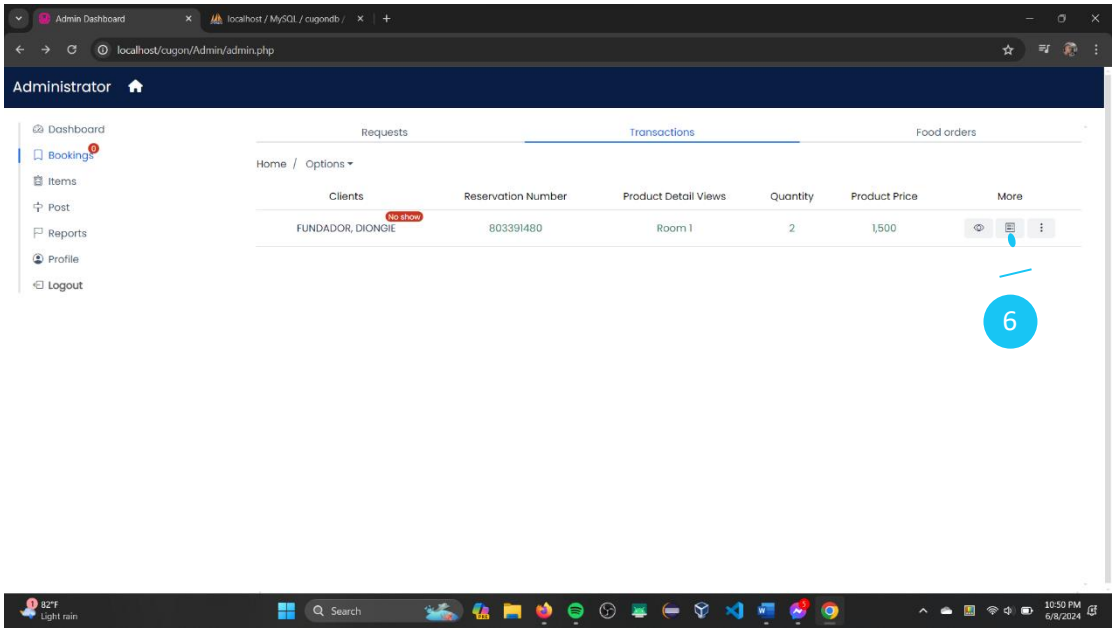
3. Click Log in to continue.



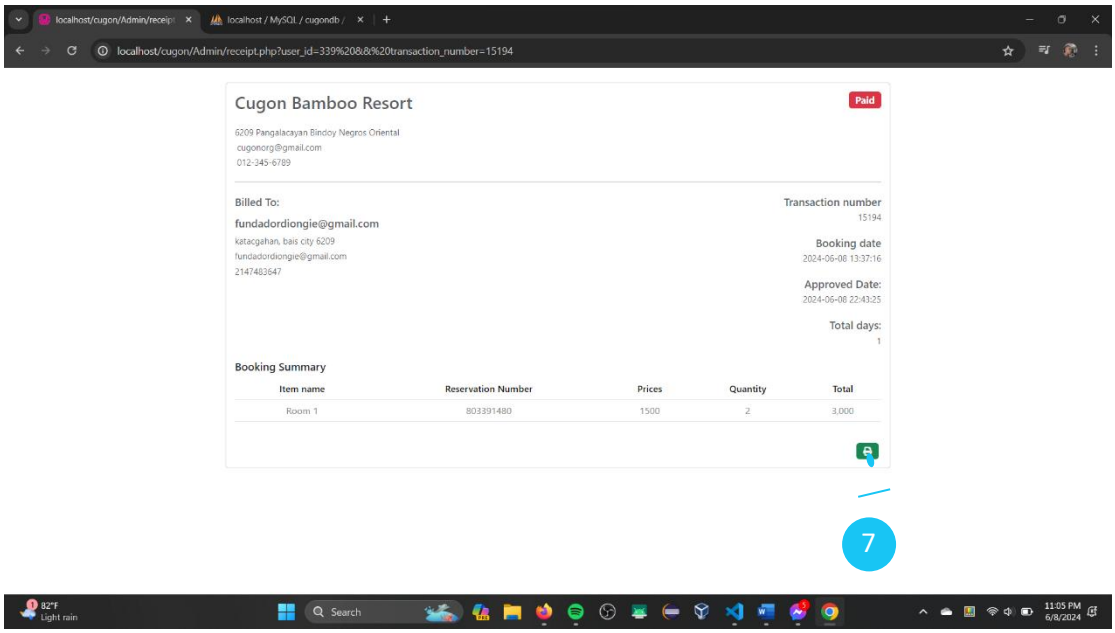
4. Click to see ***bookings***.



5. Click to ***approved*** request.



6. Click to see the ***receipt***.



7. Click to print the ***receipt***.

Appendix I. *Source Code*

For the details of the source code, please refer to the soft copy attached at the inside back cover of the hardbound copy of this project.

CURRICULUM VITAE

KERWIN P. BABOR

Brgy. Katacgahan Bais City Negros Oriental
09945954769
kerwinbabor@gmail.com

PERSONAL INFORMATION

Date of Birth:	OCTOBER 25 2001
Place of Birth:	Katacgahan bais city
Mother's Name:	Lisa P. Babor
Father's Name:	Frankie I. Babor
Religion:	Roman Catholic
Citizenship:	Filipino

EDUCATIONAL ATTAINMENT

Tertiary:	Negros Oriental State University Campus 1 Bais City Negros Oriental
Secondary:	Bais City National Highschool Bais City Negros Oriental
Primary:	Katacgahan Elementary School Katacgahan, Bais City, Negros Oriental

Diongie C. Fundador

Brgy. Sundo-an Manjuyod, Negros Oriental
09319158016
fundadordiongie@gmail.com

PERSONAL INFORMATION

Date of Birth:	August 25, 1998
Place of Birth:	Sundo-an Manjuyod
Mother's Name:	Virgie C. Fundador
Father's Name:	Dionecito E. Fundador
Religion:	INC
Citizenship:	Filipino

EDUCATIONAL ATTAINMENT

Tertiary:	Negros Oriental State University Campus 1 Bais City Negros Oriental
Secondary:	Manjuyod National High School Manjuyod, Negros Oriental
Primary:	Manjuyod Elementary School Manjuyod, Negros Oriental

Gino Q. Malco

Brgy. Sta. Cruz Viejo, Tanjay City, Negros Oriental
09289567523
Malco.gino22@gmail.com

PERSONAL INFORMATION

Date of Birth:	May 22, 2001
Place of Birth:	NOPH, Dumaguete City
Mother's Name:	Jessica Q. Malco
Father's Name:	Antonio T. Malco Sr.
Religion:	Roman Catholic
Citizenship:	Filipino

EDUCATIONAL ATTAINMENT

Tertiary:	Negros Oriental State University Campus I, Bais City, Negros Oriental
Secondary:	Lourdes Ledesma Del Prado Memorial National High School Tanjay City, Negros Oriental
Primary:	Sta. Cruz Central School Tanjay City, Negros Oriental

Mary Joyce F. Candido

Brgy. Sab-ahan Negros Oriental
09975240254
Maryjoycecandido2002@gmail.com

PERSONAL INFORMATION

Date of Birth:	September 14, 2002
Place of Birth:	Bayambang Pangasinan
Mother's Name:	Jocelyn Fernandez
Father's Name:	Emerito Candido
Religion:	Roman Catholic
Citizenship:	Filipino

EDUCATIONAL ATTAINMENT

Tertiary:	Negros Oriental State University Campus 1 Bais City Negros Oriental
Secondary:	Dodong Escaño Memorial High School Sab-ahan Negros Oriental
Primary:	Lowason Elementary School Sab-ahan Negros Oriental

Myca C. Capingcot

Brgy. Tanglad, Manjuyod, Negros Oriental
09605111037

PERSONAL INFORMATION

Date of Birth:	August 30, 2000
Place of Birth:	Manjuyod, Negros Oriental
Mother's Name:	Juliet C. Capingcot
Father's Name:	Roger A. Capingcot
Religion:	Iglesia Filipina Independiente
Citizenship:	Filipino

EDUCATIONAL ATTAINMENT

Tertiary:	Negros Oriental State University Campus I, Bais City, Negros Oriental
Secondary:	Bais City National High School Bais City, Negros Oriental
Primary:	Tanglad Elementary School Manjuyod, Negros Oriental